AEGIS: Matrix-Free Operator — Audit, Certify, and Defend Graph Neural Networks

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Abstract

Deploying a graph neural network safely takes two things that current tools provide only in isolation: a map of which edge perturbations would break its predictions, and a robustness guarantee a practitioner can act on. AEGIS delivers both from a single matrix-free object, the *constrained sensitivity matrix* S_c, which records how strongly each edge can move the model's equilibrium prediction and scales to $N{=}7{,}650$ nodes on one GPU. From this one object come three capabilities. AEGIS *audits*: its leading singular direction, column norms, and node margins give an optimal attack, an edge-vulnerability ranking, and per-node safe radii in one pass. It *certifies*: AEGIS-Conformal bounds in closed form how far a bounded perturbation can shift the conformity scores, turning split conformal prediction into a distribution-free guarantee that holds at the nominal level under the very attack it certifies, with no Monte-Carlo sampling. This certificate stays non-vacuous where randomized smoothing degenerates to the full label set at $10^4\times$ the cost, and for contractive models a closed-form safe radius adds a deterministic guarantee that the empirical breaking budget exceeds by $\mathbf{2}$–$\mathbf{9}\times$. It *defends*: penalizing the leading sensitivity trades clean accuracy for certified robustness. Empirically, this continuous sensitivity predicts discrete attack damage across 6 datasets, 7 architectures, and 4 domains (rank correlation $\tau{=}0.99$), and one analytic query inflicts $\mathbf{74}$–$\mathbf{156}\times$ the per-query damage of a 50-step PGD attack.

1 Introduction

Graph neural networks are now deployed where structural errors carry real consequences: fraud accounts evade detection via perturbed transaction edges (Zügner, Akbarnejad, and Günnemann 2018), drug-interaction models misfire under perturbed molecular graphs (Dai et al. 2018), and power grids can miss contingencies when their graph is structurally fragile (Nakiganda et al. 2023). A practitioner faces a question current tools leave unanswered: *which edges, if perturbed, would cause predictions to fail, and by how much?*

Each existing thread maps only part of these *adversarial fault lines*. Structural attacks (Zügner, Akbarnejad, and Günnemann 2018; Zügner and Günnemann 2019a; Geisler et al. 2021; Wu et al. 2019) rank edges by surrogate gradients, but return no continuous direction and no per-node budget. Smoothing (Bojchevski, Klicpera, and Günnemann 2020; Bojchevski and Günnemann 2019; Schuchardt et al. 2023, 2021) and deterministic certifiers (Li and Wang 2025; Zügner and Günnemann 2019b) return per-node certificates, but neither an edge ranking nor a direction. Robust-architecture defenses (Zhu et al. 2019; Zhang and Zitnik 2020; Jin et al. 2020) harden models without surfacing which edges are vulnerable, and critiques (Mujkanovic et al. 2022; Gosch et al. 2023) catalogue the resulting evaluation pitfalls. No prior method audits, certifies, and defends in a single matrix-free pass. AEGIS (Adversarial Evaluation of Graph Integrity via Sensitivity) does, from one object: the constrained sensitivity matrix S_c (section 3). The idea is simple: S_c records how strongly each edge can push the equilibrium prediction, so its leading singular direction, column norms, and per-node margins deliver the three diagnostics from one randomized SVD, and the same bound upgrades each per-node radius into a distribution-free certificate (section 4.1). Two scope notes: the closed-form break characterisation (Theorem 1) holds for contractive implicit GNNs, while S_c extends to any GNN with continuous edge-weight message passing (section 5.1); the threat model is edge deletion and weight perturbation (section 2).

Contributions. (1) Constrained sensitivity operator. S_c specialises equilibrium IFT sensitivity (Koh and Liang 2017; Gould, Hartley, and Campbell 2021) to *structural* edge perturbations via P_c; one matrix-free query reads the SVD-optimal direction, per-edge rankings, and per-node radii together, where no prior object yields all three (matching a dense σ_1 to 0.03% at $N{=}200$, scaling to $N{=}7{,}650$). **(2) Certification from sensitivity.** The same bound yields two guarantees: AEGIS-Conformal, a *distribution-free* certificate with coverage $\geq 1{-}\alpha$ over the ε-ball that holds at the nominal level under worst-case attack and stays non-vacuous where the deterministic radius is thin (section 4.1); and, for contractive models, a constant-factor two-sided characterisation of the breaking point $\varepsilon_{\mathrm{crit}}{=}(1{-}\kappa)/\left\|W\right\|_2$, whose norm certificate under-states the true break by 2–9$\times$ (10 seeds). **(3) Coupled defense.** Penalizing $\sigma_1(S_c)$ trades clean accuracy for certified robustness, so the operator that finds the worst attack also tunes the defense (section 5.3). **(4) Empirical evaluation.** 6 datasets, 7 architectures, 4 domains (390 runs): four-quadrant attack comparison, head-to-head vs. GR-BCD/PR-BCD (Geisler et al. 2021) and complementary positioning against AGNNCert (Li and Wang 2025) (appendix F.2), continuous-to-discrete transfer (fig. 5), and a

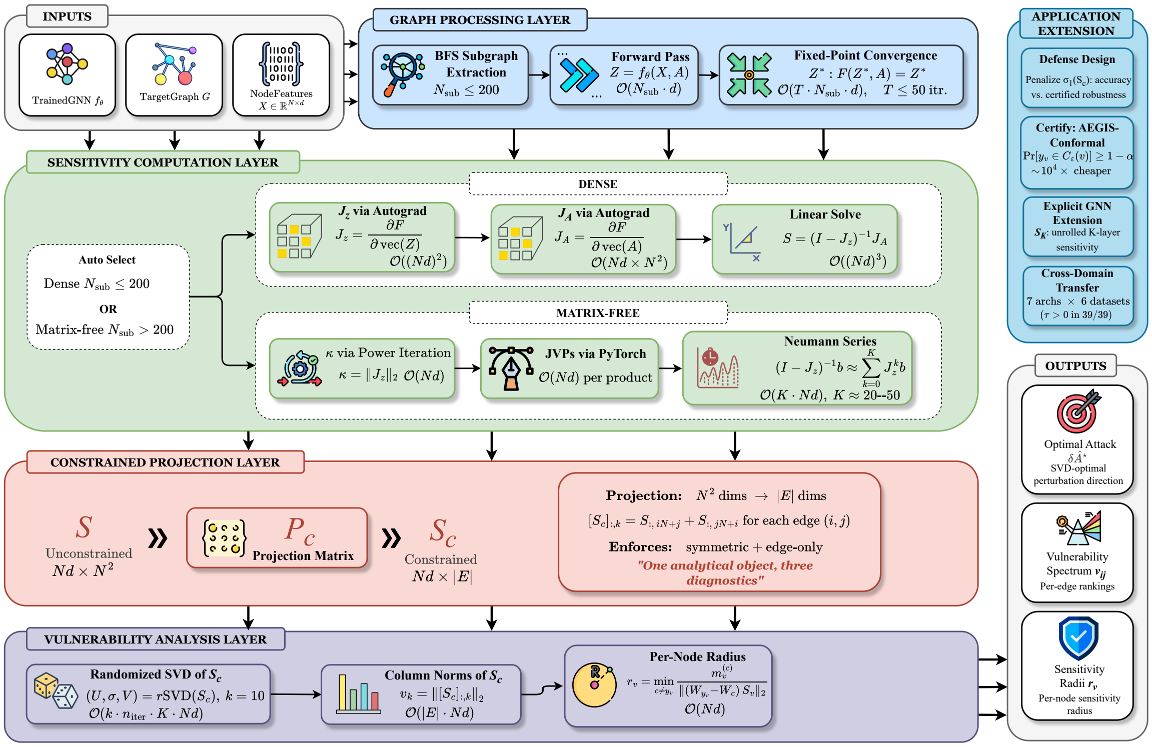


Figure 1: The four-stage AEGIS pipeline (section 3): (1) graph processing (BFS subgraph or full-graph pass); (2) sensitivity (dense Jacobian for $N \le 200$, else matrix-free Neumann/JVP); (3) constrained projection $N^2 \to |E|$ via P_c; (4) analysis (randomised SVD, S_c column norms, per-node radii).

fraud audit (section 6).

2 Preliminaries and Threat Model

Let $G = (V, E, A)$ be an undirected graph with $N = |V|$ nodes, adjacency $A \in \mathbb{R}^{N\times N}$, features $X \in \mathbb{R}^{N\times d_{\text{in}}}$, and symmetric normalized adjacency $\hat{A} = D^{-1/2}(A+I)D^{-1/2}$. Throughout, ϕ is the activation, $\sigma_i(\cdot)$ the i-th singular value, $\text{vec}(\cdot)$ column-major vectorization. The sensitivity matrices (S, S_c, S_K, S_v) and governing scalars ($\kappa{=}\|J_z\|_2$, $\rho(J_z)$, η, $\varepsilon_{\text{crit}}$, r_v) are defined where they first appear.

IGNN and IFT. AEGIS analyses models that settle to an equilibrium, because that fixed point is what exposes structural sensitivity in closed form. A DEQ-GNN (Bai, Kolter, and Koltun 2019; Bai, Koltun, and Kolter 2020) iterates a weight-tied F_θ to a fixed point; the IGNN instantiation (Gu et al. 2020) uses

$$F(Z, A) = \phi(\hat{A} Z W^\top + X_{\text{proj}}), \tag{1}$$

with $W \in \mathbb{R}^{d\times d}$ spectral-normalized (Miyato et al. 2018) so $\kappa \le \left\|\hat{A}\right\|_2 \|W\|_2 < 1$, ReLU ϕ, and a learned projection X_{proj}. Training uses implicit differentiation through the fixed point (Gould, Hartley, and Campbell 2021; Fung et al. 2022; Bai, Koltun, and Kolter 2021), and a linear head $f(z^*){=}W_{\text{cls}} z^* {+} b$ gives class predictions. At equilibrium, $G(z^*, A){=}z^* {-} F(z^*, A){=}0$ defines z^* as an implicit function of A, with IFT sensitivity

$$\frac{\partial z^*}{\partial p} = (I - J_z)^{-1} \left.\frac{\partial F}{\partial p}\right|_{z^*}. \tag{2}$$

Equation (2) is the lever AEGIS pulls: the closed-form first-order change in z^* under any parameter p. The resolvent $(I - J_z)^{-1}$ acts as a gain that amplifies the perturbation by at most $1/(1 - \kappa)$, so the closer the model drifts to losing contractivity ($\kappa \to 1$), the more violently its equilibrium can move; the phase transition of section 4 makes this precise, with one refinement: the pseudospectral index $\eta{=}\left\|(I - J_z)^{-1}\right\|_2 (1 - \rho) \ge 1$ (Trefethen and Embree 2005) measures how far the true gain sits above its spectral-radius estimate, equalling 1 when J_z is normal.

Threat model. A white-box attacker with full access to the trained IGNN perturbs the normalized adjacency

$$\hat{A}' = \hat{A} + \delta\hat{A}, \quad \left\|\delta\hat{A}\right\|_F \le \varepsilon, \tag{3}$$

with $\delta\hat{A}$ symmetric ($\delta\hat{A}{=}\delta\hat{A}^\top$, preserving the undirected graph), edge-only ($[\delta\hat{A}]_{ij}{=}0$ for $(i,j) \notin E$, so no new edges), and continuous in $\mathbb{R}$. AEGIS targets edge deletion and weight perturbation; insertion attacks (Nettack-class (Zügner, Akbarnejad, and Günnemann 2018; Zügner and Günnemann 2019a) and node injection (Sun et al. 2020)) would enlarge the basis to $\bar{E} \supseteq E$, a natural follow-up that leaves the S_c construction intact, and discrete deletions connect to the continuous model through Proposition 3.

Unlike input-feature perturbation (El Ghaoui et al. 2021; Revay, Wang, and Manchester 2020), structural perturbation changes the operator itself: z^* depends on A through both $J_z{=}\mathrm{diag}(\phi')(\hat{A}\otimes W)$ and $J_A{=}\partial F/\partial\mathrm{vec}(A)$, motivating S_c (section 3).

3 The AEGIS Operator Construction

The previous section isolated the two Jacobians, J_z and J_A, through which the adjacency moves the equilibrium. AEGIS packages them into one operator, the constrained sensitivity matrix S_c, and reads every diagnostic off it in a single computation, including $\varepsilon_{\mathrm{crit}}$ and the regime characterisation of Theorem 1 for IGNN-class operators. None of this builds S_c explicitly: the pipeline only applies it to vectors, never materialising $S \in \mathbb{R}^{Nd\times N^2}$ or the resolvent. A dense path ($N \leq 200$) takes an exact SVD; a matrix-free path ($N > 200$, validated to $N{=}7{,}650$) uses JVP/VJP queries and a randomized SVD. Figure 1 shows the four stages, algorithm 1 the routing.

Algorithm 1: AEGIS matrix-free vulnerability analysis. S_c is queried but never formed; the dense fallback ($N \leq 200$) builds S_c and runs an exact SVD.

Require: F, fixed point z^*, $\hat{A}$, edge set E, budget ε, rank k, Neumann tolerance tol; (opt.) head W, margins $m_v^{(c)}$.
Ensure: $\sigma_1(S_c)$, $\delta\hat{A}^\star$, $\{v_{ij}\}$, $\{r_v\}$, $\varepsilon_{\mathrm{crit}}$.

$\kappa \leftarrow \|J_z\|_2$; $K \leftarrow \lceil \log(1/\mathrm{tol})/\log(1/\kappa)\rceil$
$S_c\, v := \left(\sum_{j=0}^{K} J_z^j\right) J_A\, P_c\, v$ {matrix-free}
$(\sigma_1, v_1) \leftarrow \mathrm{rSVD}(S_c, k)$ (Halko, Martinsson, and Tropp 2011)
$\delta\hat{A}^\star \leftarrow \varepsilon\, \mathrm{sym}(P_c v_1) / \|\mathrm{sym}(P_c v_1)\|_2$
$v_{ij} \leftarrow \|S_c\, e_{ij}\|_2$ for each $(i,j) \in E$
$r_v \leftarrow \min_{c\neq y_v} m_v^{(c)} / \|(W_{y_v} - W_c)\, S_{c,v}\|_2$ {head only}
$\varepsilon_{\mathrm{crit}} \leftarrow (1-\kappa)/\|W\|_2$ {Theorem 1}
return $(\sigma_1,\ \delta\hat{A}^\star,\ \{v_{ij}\},\ \{r_v\},\ \varepsilon_{\mathrm{crit}})$

The constrained sensitivity operator. S_c is the full sensitivity S cut down to the perturbations the threat model permits: one column per existing edge, symmetrised so that touching edge (i,j) also touches (j,i). The edge basis has unit Frobenius norm, so $\sigma_1(S_c)$ matches the budget $\left\|\delta\hat{A}\right\|_F{=}\varepsilon$; the $N^2 \to |E|$ restriction P_c is the standard duplication-matrix reduction (Magnus and Neudecker 2019) onto the edge-supported symmetric subspace. One randomized SVD of S_c yields the three audit readings together: the optimal attack direction (Proposition 1), the per-edge ranking $v_{ij}{=}\|[S_c]_{:,k}\|_2$, and the per-node radius r_v (Proposition 2). The same operator then *certifies* the model (section 4.1) and *defends* it (section 5.3).

Proposition 1 (Maximally Sensitive First-Order Structural Perturbation)**.** *The perturbation δA^* maximizing $\|\Delta z^*\|$ to first order subject to $\|\delta A\|_F \leq \varepsilon$ is $\delta A^* = \varepsilon \cdot \mathrm{reshape}(v_1, N\times N)$, where v_1 is the leading right singular vector of S, with shift $\varepsilon\cdot\sigma_1(S)$. Restricted to the threat model's feasible set (symmetric, edge-supported; section 2) the maximizer is instead the leading right singular vector of the constrained S_c in the edge basis $\{b_k\}$, with shift $\varepsilon\,\sigma_1(S_c){\leq}\varepsilon\,\sigma_1(S)$; the unconstrained $\mathrm{reshape}(v_1)$ is generally infeasible.*

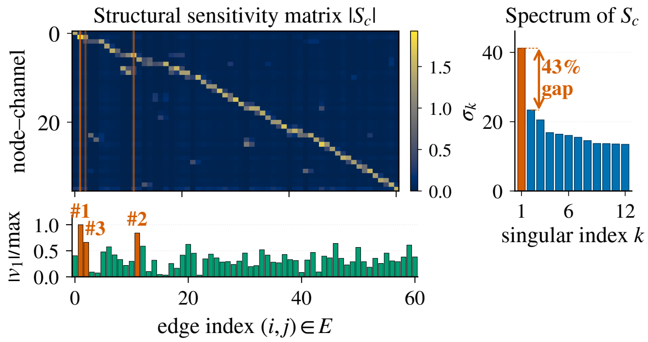


Figure 2: $|S_c|$ on a 50-node Cora ego-graph (IGNN, $N{=}50$, $d{=}16$, $|E|{=}61$, seed 0). Vermilion: top-3 edges by $|v_1|$ (SVD-optimal direction). Sidebar: singular spectrum ($\sigma_1{=}41.2$, 43% gap to σ_2).

This direction maximises hidden-state shift rather than label change, so it pinpoints the most dangerous way to spend the budget rather than certifying a global margin, yet still aligns empirically with classification damage (section 5.1).

Matrix-free operator. Scale rests on never building S_c: applying it to a vector, $S_c v = (I{-}J_z)^{-1} J_A P_c v$, costs $O(Nd)$ memory instead of the $O((Nd)^2)$ a dense S_c needs. We evaluate the resolvent by a truncated Neumann series whose depth adapts to κ ($K{\in}[20, 50]$ for $\kappa{<}0.8$), each term a forward-mode JVP with P_c applied implicitly; the dense path ($N \leq 200$) instead materialises J_z, J_A directly. A randomized SVD (Halko, Martinsson, and Tropp 2011) returns the leading (σ_1, v_1), symmetrised so the recovered attack stays in the threat model. One query suffices because the singular spectrum is well separated (gap 0.39–0.50; fig. 2), which is what lets a single rSVD match iterative attackers (section 5.1); removing the $O((Nd)^3)$ dense solve lifts the previous $N{\approx}300$ ceiling (appendix F.4).

4 Adversarial Equilibrium Theory

Section 3 read three diagnostics off S_c; we now justify them, starting with the safety boundary. The picture is a phase transition in ε: while ε is small the model stays contractive and its prediction moves boundedly with the graph; as ε nears the critical budget $\varepsilon_{\mathrm{crit}}$ the equilibrium grows arbitrarily sensitive; past $\varepsilon_{\mathrm{crit}}$ contraction fails and the prediction can break. Theorem 1 makes this precise and puts $\varepsilon_{\mathrm{crit}}$ in closed form.

Theorem 1 (Vulnerability Characterization for Contractive Implicit GNNs)**.** *Let $F_\theta(Z, \hat{A}) = \phi(\hat{A}ZW^\top + X_{\mathrm{proj}})$ be an IGNN-class operator satisfying:*

(A1) ϕ is ReLU or any 1-Lipschitz activation; nonsmoothness at zero is handled via the conservative IFT (Bolte and Pauwels 2021) (proof below).
(A2) $\|W\|_2 \leq c$ via spectral normalization.
(A3) $\|J_z\|_2 \leq \kappa < 1$ verified post-training (κ=0.14–0.59 across our suite; table 1). (A2) does not imply (A3); we report the trained κ so practitioners can audit, and AEGIS *still returns rankings and attacks if (A3) fails, its guarantees then scoped to the contractive regime.*

Define the critical budget

$$\varepsilon_{\mathrm{crit}} = \frac{1-\kappa}{\|W\|_2}. \tag{4}$$

Then structural perturbations $\delta\hat{A}$ with $\left\|\delta\hat{A}\right\|_F = \varepsilon$ exhibit three regimes:

***(a) Subcritical** ($\varepsilon < \varepsilon_{\mathrm{crit}}$): The perturbed operator remains contractive with a unique fixed point satisfying*

$$\|\Delta z^*\|_F \leq \sigma_1(S) \cdot \varepsilon + O(\varepsilon^2), \tag{5}$$

where $S = (I - J_z)^{-1} J_A$ is the structural sensitivity matrix and $\sigma_1(S) \leq \|J_A\|_{\mathrm{op}} / (1-\kappa)$.

***(b) Critical** ($\varepsilon \to \varepsilon_{\mathrm{crit}}$): The resolvent obeys $\left\|(I-J_z')^{-1}\right\|_2 \geq 1/\min_i |1-\lambda_i(J_z')|$ and blows up when an eigenvalue of J_z' approaches 1 (a vanishing spectral gap to 1, not $\|J_z'\|_2 \to 1$). The rate is $\Omega(1/(\varepsilon_{\mathrm{crit}}-\varepsilon))$ for normal J_z' with a dominant Perron eigenvalue; in general* $\varepsilon_{\mathrm{crit}}$ lower-bounds *the divergence threshold, the slack being the nonnormality η ($\eta \leq 2.47$ for ReLU; Observation 1 and Remark 1).*

***(c) Supercritical** ($\varepsilon \geq \varepsilon_{\mathrm{crit}}$): The contraction certificate no longer applies ($\|J_z'\|_2$ may exceed 1); $\varepsilon < \varepsilon_{\mathrm{crit}}$ is a sufficient safety boundary that trained IGNNs sit well inside (2–4$\times$ spectral-radius margin to ρ=1, appendix F.4).*

Proof in section B.1. The subcritical bound is a conservative IFT (Bolte and Pauwels 2021) with Neumann $\left\|(I-J_z)^{-1}\right\|_2 \leq 1/(1-\kappa)$; the critical rate follows from the resolvent identity.

The closed-form $\varepsilon_{\mathrm{crit}}$ is only *sufficient*; the data-dependent S_c tightens it by 7–14$\times$. The slack is the Neumann-versus-spectral-radius gap, the pseudospectral index η, which for symmetric $\hat{A}$ is controlled by W alone ($\eta \in [1.19, 2.47]$ for ReLU; Observation 1 and Remark 1).

Theorem 1(a) certifies safety *below* $\varepsilon_{\mathrm{crit}}$; the same operator also bounds the boundary from *above*, through the nonnormality $g_W = \|W\|_2 / \rho(W) \geq 1$ of W.

Theorem 2 (Constant-factor two-sided boundary, all-active)**.** *Under* (A1)–(A3) *with $\hat{A}$ symmetric and $\varepsilon_{\mathrm{crit}} > 0$, the all-active contraction boundary $\varepsilon_{\mathrm{br}}$ (the smallest feasible $\left\|\delta\hat{A}\right\|_F$ at which $\rho(\hat{A}' \otimes W) \geq 1$) obeys*

$$\varepsilon_{\mathrm{crit}} \;\leq\; \varepsilon_{\mathrm{br}} \;\leq\; \tfrac{C}{\beta}\,\varepsilon_{\mathrm{crit}}, \qquad C = g_W\,\tfrac{1+\kappa}{1-\kappa}, \tag{6}$$

with C dimension-free and computable post-training, and $\beta \in (0,1]$ the edge-supported Frobenius mass of $\hat{A}$'s Perron mode (positive for any connected graph). The upper side is attained by a rank-one critical-driving attack aligned with that mode; the bracket collapses to equality iff g_W=1 and β=1.

The bracket constant is loose ($\sim$10–16$\times$), but it exposes the coupling: one operator S_c supplies the certified budget $\varepsilon_{\mathrm{crit}}$, the matching attack that approaches it, and, through $\sigma_1(S_c)$, the defense. Empirically the norm certificate understates the true break by **2–9**$\times$ (10 seeds); proof and nonlinear extension in appendix D.

The per-node radius r_v (section 3) carries a first-order safety guarantee.

Proposition 2 (Per-Node First-Order Sensitivity Radius)**.** *For a linear head $f(z)$=Wz+b and a node v classified as y_v with positive margins $m_v^{(c)}$=$f_{y_v}(z_v^*)-f_c(z_v^*)>0$ to every competitor $c \neq y_v$, the first-order sensitivity radius*

$$r_v = \min_{c \neq y_v} \frac{m_v^{(c)}}{\|(W_{y_v} - W_c)\, S_v\|_2} \tag{7}$$

preserves the classification of v for any $\left\|\delta\hat{A}\right\|_F < r_v$, where S_v are the block-rows of S at node v; substituting $S_{c,v}$ gives the tighter constrained radius $r_v^{(c)} \geq r_v$. Minimizing over all *competitors is the distance to the nearest first-order boundary, so the bound holds even when the runner-up changes.*

Proof in section B.3. A Cauchy–Schwarz bound on each per-competitor margin change gives r_v as the smallest margin divided by two amplification factors (decision-boundary and equilibrium sensitivity); high-margin, low-sensitivity nodes get large radii.

4.1 AEGIS-Conformal: A Distribution-Free Certificate

The radius r_v of eq. (7) is a first-order threshold: locally tight, but liable to be violated at larger ε. We now upgrade it to a distribution-free guarantee, one that is sound under an exchangeability condition and gated by an explicit coverage test (section E). The starting point is split conformal prediction, which returns a set $C(v)$ with $\Pr[y_v \in C(v)] \geq 1-\alpha$ on clean inputs from a *conformity score*; we report the two standard choices, *TPS* (the softmax probability of the candidate label; Sadinle, Lei, and Wasserman 2019) and *APS* (its adaptive cumulative variant; Romano, Sesia, and Candès 2020), where APS yields slightly larger sets and TPS can abstain with an empty one. Making this hold *robustly* over the ball $\left\|\delta\hat{A}\right\|_F \leq \varepsilon$ requires bounding the worst-case shift in the conformity scores, and that is exactly what the same S_c sensitivity delivers in closed form: a curvature-corrected bound $L_1^c\varepsilon + C_v\varepsilon^2$ that replaces the $\sim 10^4$-sample smoothing the task would otherwise demand. Substituting it into the binary-certificate split conformal of Zargarbashi, Antonelli, and Bojchevski (2023) yields a robust set with

$$\Pr\bigl[y_v \in C_\varepsilon(v)\bigr] \;\geq\; 1-\alpha \qquad \text{for all } \left\|\delta\hat{A}\right\|_F \leq \varepsilon. \tag{8}$$

Section 5.2 validates this certificate empirically, checking both nominal-level coverage and the coverage gate under the worst-case attack, and section 5.3 returns to the coupled $\sigma_1(S_c)$ defense.

4.2 Transfer to Discrete Edits and Explicit GNNs

Two bridges carry S_c from the continuous, implicit setting to the discrete, architecture-general one practitioners face; both are proved in section C. The first reads a discrete edge-removal ranking off the continuous score. Deleting an edge is, to first order, a continuous perturbation whose size is the edge's own weight, so its damage should track the score S_c already assigns it.

Proposition 3 (Continuous-to-discrete transfer)**.** *Under* (A1)–(A3)*, let edge* k *have weight* w_k *and score* $v_k = \|[S_c]_{:,k}\|_2$*, and let* d_k *be the true damage of removing it. If every single-edge removal is subcritical, then*

$$d_k = w_k v_k + R_k, \qquad |R_k| \le \tfrac{L_J}{2(1-\kappa)^2}\, w_k^2, \tag{9}$$

with curvature constant $L_J \le \|W\|_2^2 \|z^*\|$*. The edge-weighted score* $w_k v_k$ *therefore preserves the brute-force removal order whenever the score gap between two edges beats this quadratic remainder.*

The signal is $O(w_k)$ and the error only $O(w_k^2)$, so the ranking is reliable for the small-weight edges of sparse graphs; empirically $w_k v_k$ reaches $\tau = +\mathbf{0.996}$ against brute-force N-1 on full-graph Amazon Photo (section 5.1). The second bridge drops the implicit-model assumption: an explicit K-layer GNN has no fixed point, but unrolling its layers yields the same kind of structural Jacobian.

Proposition 4 (Explicit K-layer GNNs)**.** *For* $Z_K = g_K \circ \cdots \circ g_1(X, A)$ *with per-layer Jacobians* $J_z^{(k)} = \partial g_k / \partial Z_{k-1}$ *and* $J_A^{(l)} = \partial g_l / \partial \mathrm{vec}(A)$*, the unrolled sensitivity*

$$S_K = \frac{\partial\, \mathrm{vec}(Z_K)}{\partial\, \mathrm{vec}(A)} = \sum_{l=1}^{K} \Big(\prod_{k=l+1}^{K} J_z^{(k)} \Big) J_A^{(l)} \tag{10}$$

replaces S *throughout: the constrained operator* S_c*, the per-node radius* r_v*, and the first-order shift bound* $\|\Delta Z_K\|_F \le \sigma_1(S_K)\, \|\delta A\|_F$ *all carry over unchanged.*

So six of our seven architectures inherit the full audit/certify/defend pipeline from S_K; only the closed-form boundary $\varepsilon_{\mathrm{crit}}$ stays restricted to the contractive implicit subclass.

5 Experiments

Our evaluation follows the three capabilities of section 3: AEGIS *audits* (section 5.1), *certifies* (section 5.2), and *defends* (section 5.3); a deployed fraud detector (section 6) then closes the loop on real data.

Setup. We evaluate AEGIS on 6 datasets across 4 domains, 10 seeds each (PyTorch (Paszke et al. 2019); ± are seed std). The IGNN (Gu et al. 2020) of eq. (1) uses $W \in \mathbb{R}^{64\times 64}$ spectral-normalised (Miyato et al. 2018) with Adam (lr 0.01). Datasets: Cora, Citeseer, Pubmed (Sen et al. 2008), Amazon Photo (Shchur et al. 2018), WikiCS (Mernyei and Caron 2020), Amazon Fraud (Dou et al. 2020); baselines: smoothing (Gaussian σ=0.01), Mettack, GR-BCD/PR-BCD (Geisler et al. 2021). Non-IGNN architectures appear only in fig. 5 and table 9. Full tables, ablations, and timing in appendix F; reproducibility in appendix H.

5.1 Audit: One Query Finds the Worst Attack

Two prerequisites must hold: the trained models sit in the contractive regime, and AEGIS's perturbations beat chance. Table 1 confirms both, verifying (A3) on every benchmark (κ<1) and reporting per-dataset $\varepsilon_{\mathrm{crit}}$ with a 3.2–4.1× attack advantage over random. Unless stated otherwise, IGNN runs on 50-node BFS ego-subgraphs around the highest-degree node and reports *local* vulnerability; graph-level assessment uses the matrix-free pipeline (appendix F.4), and all bounds use $\kappa = \|J_z\|_2$.

Table 1: Per-dataset IGNN characterisation (10 seeds). $\kappa = \|J_z\|_2$ verifies (A3); $\varepsilon_{\mathrm{crit}} = (1-\kappa)/\|W\|_2$ is the local critical budget. AtkAdv: AEGIS/random damage. Cert: subgraph nodes certified first-order-safe at ε=0.05 (r_v>0.05).

Dataset	Acc	κ	$\varepsilon_{\mathrm{crit}}$	AtkAdv	Cert
Cora	77.5$_{\pm 1.7}$	.33$_{\pm .14}$	.66$_{\pm .15}$	3.6$_{\pm .6}$	.83$_{\pm .10}$
Citeseer	66.0$_{\pm 0.7}$	.59$_{\pm .02}$	.41$_{\pm .02}$	4.1$_{\pm .5}$	.94$_{\pm .05}$
Pubmed	78.9$_{\pm 0.7}$	.41$_{\pm .11}$	.59$_{\pm .11}$	3.2$_{\pm .4}$	.76$_{\pm .18}$
Amazon	94.8$_{\pm 0.3}$	.14$_{\pm .02}$	.86$_{\pm .02}$	3.8$_{\pm .2}$	.86$_{\pm .08}$
WikiCS	77.9$_{\pm 0.4}$	.34$_{\pm .04}$	.66$_{\pm .04}$	3.8$_{\pm .4}$	.78$_{\pm .04}$

Table 2: One-query audit vs. iterative attackers: equilibrium damage at ε=0.10 (IGNN, 50-node subgraph, 10 seeds). Cls-PGD: classification-loss PGD; Shift-PGD: IFT-gradient PGD (solver validation, not an independent baseline).

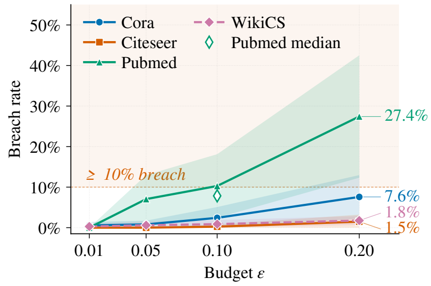
Dataset	AEGIS	Cls-PGD	Shift-PGD	Random
Cora	3.70$_{\pm .79}$	2.51$_{\pm .48}$	3.05$_{\pm .70}$	0.97$_{\pm .25}$
Citeseer	4.63$_{\pm .71}$	2.97$_{\pm .42}$	3.35$_{\pm .57}$	1.01$_{\pm .15}$
WikiCS	2.10$_{\pm .22}$	0.67$_{\pm .11}$	1.93$_{\pm .20}$	0.53$_{\pm .09}$

One query, not a search. AEGIS spends its whole budget in one analytic query: the leading singular direction of S_c is, by construction, the perturbation that moves the equilibrium most (Proposition 1), with no optimisation loop. The worry with a closed-form attack is that it exploits the victim's gradients rather than a real fault line; three checks rule this out. *It transfers:* a direction from a *separately trained* surrogate (no shared gradients) still recovers 99% of the one-query damage (cos =0.99), against 44±4% for a 512-query black-box search, so the fault line belongs to the model, not to gradient access. *It is efficient:* in table 2 one query matches or beats a 50-step PGD (Madry et al. 2018) at ε=0.10, a **74–156**× advantage *per query*; only GR-BCD overtakes it, at the large budgets it targets (appendix F.2). *It is faithful:* per-edge finite differences reproduce the S_c ranking almost exactly (τ=0.999), and re-solving after every removal (the greedy ideal) closes at most 3 pp of the gap, so the one-shot ranking is the headline.

AEGIS also beats degree, edge-betweenness, and spectral rankings at ε=0.01 (+6–148% AtkAdv) and inflicts 3–10× more equilibrium damage than Mettack (Zügner and Günnemann 2019a) in the early-warning regime ($\mathbf{149/150}$ wins, $p<10^{-43}$); discrete-removal curves vs. label-aware Greedy are in fig. 4 (GR-BCD/PR-BCD in appendix F.2). The first-order bounds match actual shifts to within 1% at small budgets, loosening to ≤39% on citation graphs by ε=0.20.

Breach rates. Every breached node satisfies $\varepsilon > r_v$ across all datasets and budgets, validating r_v as a first-order screen (section 5.2); flip fractions (fig. 3) stay below 2% on Citeseer/WikiCS and reach 7.6% (Cora) and a right-skewed 27.4% (Pubmed) at ε=0.20.

Continuous-to-discrete transfer. The bridges of section 4.2 extend the construction beyond implicit models: for seven K-layer architectures (GCN, SAGE, GIN,



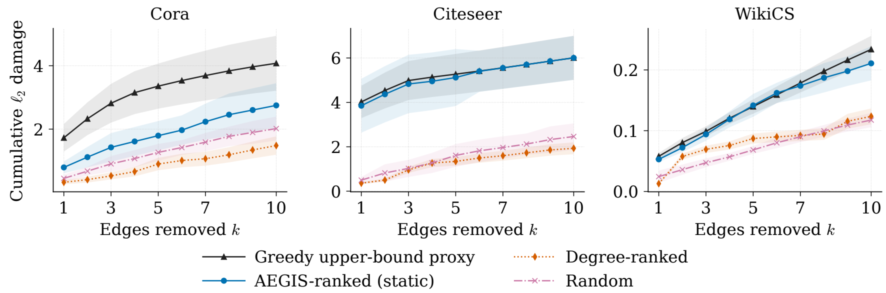
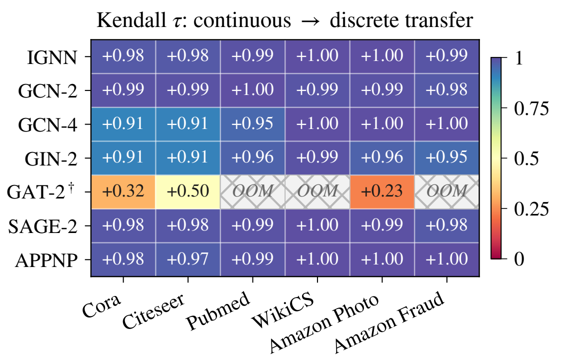


Figure 3: Breach rate (% of nodes flipped under S_c-optimal perturbation) vs. ε (10 seeds; mean, 95% CI bands).

Figure 4: Cumulative damage from sequential discrete edge removal vs. k (IGNN, 50-node subgraph, 10 seeds). AEGIS *matches* label-aware Greedy on Citeseer/WikiCS and recovers **54–67**% of its Cora damage at k=5–10 with no label access.



† *GAT-2: 2-layer attention; OOM on Pubmed, WikiCS, Amazon Fraud.*

Figure 5: Continuous-to-discrete transfer τ (10-seed means; per-cell sd 0.0003–0.041). Cross-hatch: GPU OOM. GAT-2† transfers better unweighted (appendix F.6).

APPNP, GAT† (Kipf and Welling 2017; Hamilton, Ying, and Leskovec 2017; Xu et al. 2019b; Klicpera, Bojchevski, and Günnemann 2019; Veličković et al. 2018)) we form S_K by finite differences (Proposition 4) and ask whether the continuous S_c ranking predicts the discrete brute-force N-1 removal order. Ranking by the edge-weighted score $A_{ij}v_{ij}$ (Proposition 3), fig. 5 reports Kendall τ over 6 datasets and 7 architectures (390 runs): all $\mathbf{39/39}$ cells positive, median τ=$+\mathbf{0.99}$ (p<10^{-5}), vs. 34/39 unweighted; on full-graph Amazon Photo (N=7,650, $\kappa\approx$1.00) it reaches $\tau = +\mathbf{0.996}\pm\mathbf{0.002}$. Per-architecture detail, the sweeps, and ablations are in appendices F.4 to F.6; the ablations show vulnerability is *delocalized*, so S_c localises risk for triage rather than removing it.

5.2 Certify: Distribution-Free Coverage

The certificate of section 4.1 holds empirically. Table 3 reports coverage over 10 seeds and four datasets. The crucial quantity is the *gate*: coverage retained under the worst-case AEGIS attack at magnitude ε. It sits at the nominal $1-\alpha$=0.90 at ε=0.01 and turns conservative (0.94–1.00) at ε=0.05, with sets that stay informative (∼1.0–3.5 labels) and tighten where the deterministic radius is thin, at zero Monte-Carlo samples. Two caveats: exchangeability on a single transductive graph is an assumption the gate evidences empirically, and S_c is formed densely at N=200, leaving a matrix-free pass to future work.

Table 3: AEGIS-Conformal coverage (10 seeds; Cora, Citeseer, Pubmed, WikiCS; α=0.1, target 0.90). Each ε block reports clean coverage / mean set size, then the bold **gate** (coverage under worst-case attack at ε). Classes 7/6/3/10; seed std ≤0.07; sets <1 reflect empty (abstaining) TPS sets.

Dataset	Score	ε=0.01		ε=0.05	
Cora	APS	0.90 / 1.37	**0.90**	0.89 / 1.06	**0.98**
	TPS	0.88 / 0.95	**0.90**	0.89 / 0.99	**0.98**
Citeseer	APS	0.92 / 1.50	**0.92**	0.92 / 1.35	**0.97**
	TPS	0.91 / 1.20	**0.92**	0.92 / 1.26	**0.96**
Pubmed	APS	0.91 / 1.42	**0.95**	0.92 / 1.52	**1.00**
	TPS	0.90 / 1.15	**0.92**	0.90 / 1.15	**0.98**
WikiCS	APS	0.92 / 3.70	**0.94**	0.93 / 3.62	**0.95**
	TPS	0.93 / 3.52	**0.94**	0.93 / 3.52	**0.95**

Table 4: AEGIS-Conformal vs. randomized smoothing (Cora, n=200, 10 seeds, α=0.1, APS). **frob**: matched Frobenius ball (σ=$\varepsilon/\sqrt{2|E|}$); **per_edge**: larger per-coordinate ball (σ=ε), favorable to smoothing. Cert: fraction certified; wall-clock at M=10^4 samples (AEGIS-Conformal is zero-sample).

Method	Match	ε	Cov	Set	Cert	Wall@10^4
AEGIS-Conformal	–	0.01	0.90	1.36	–	∼1 **s**
RandSmoothing	frob	0.01	1.00	7.00	0.00	7,600 s
RandSmoothing	per_edge	0.01	0.95	1.26	0.96	15,000 s
AEGIS-Conformal	–	0.05	0.90	1.05	–	∼1 **s**
RandSmoothing	frob	0.05	1.00	7.00	0.00	10,900 s
RandSmoothing	per_edge	0.05	0.99	2.39	0.77	36,300 s

Table 4 pits AEGIS-Conformal against randomized smoothing: on the matched Frobenius ball smoothing is *vacuous* (full label set, zero certified nodes), certifying only on a strictly larger per-coordinate ball, and then at **23,000–57,000**× the cost of the zero-sample S_c bound (appendix F.3).

5.3 Defend: A $\sigma_1(S_c)$ Robustness Knob

The operator that finds the worst attack also tunes the defense. Penalizing $\sigma_1(S_c)$ during training drives the dominant sensitivity down, and with it worst-case attack damage falls while the certified fraction rises (table 5). At λ=3×10^{-4} this cuts σ_1 by ∼10× and lifts the certified fraction to 0.82±0.03 for ∼4 accuracy points. The gain is genuinely

Table 5: The $\sigma_1(S_c)$ penalty trades clean accuracy for certified robustness (Cora, c=0.9 IGNN, 10 seeds). Cert (certified fraction) peaks at λ=3×10^{-3} before margin collapse.

λ	Acc	$\sigma_1(S_c)$	Cert	Attack Dmg
0	$78.1_{\pm 1.8}$	$319_{\pm 59}$	$.40_{\pm .08}$	$27.8_{\pm 6.3}$
3×10^{-4}	$73.9_{\pm 0.8}$	$32.6_{\pm 1.7}$	$.82_{\pm .03}$	$3.1_{\pm 0.2}$
1×10^{-3}	$69.0_{\pm 0.7}$	$10.7_{\pm 0.6}$	$.89_{\pm .02}$	$1.1_{\pm 0.1}$
3×10^{-3}	$61.9_{\pm 0.6}$	$3.9_{\pm 0.2}$	$.92_{\pm .02}$	$0.4_{\pm 0.0}$
1×10^{-2}	$56.4_{\pm 0.6}$	$0.8_{\pm 0.1}$	$.86_{\pm .04}$	$0.1_{\pm 0.0}$

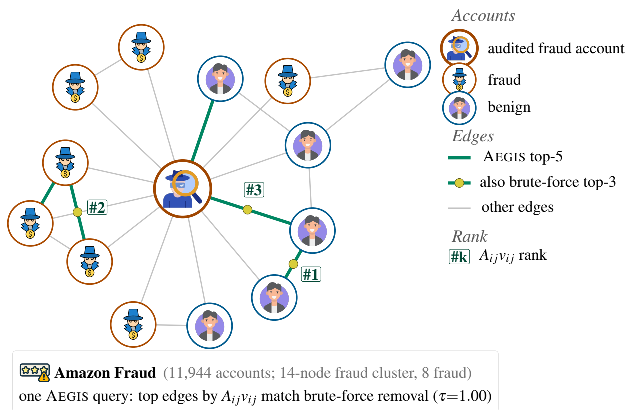


Figure 6: AEGIS auditing an IGNN fraud detector on an Amazon Fraud cluster (Dou et al. 2020). Icons mark fraud / benign accounts; the audited account is the hub. Green edges are the top-5 fault lines by $A_{ij}v_{ij}$; yellow dots mark the three also in the brute-force single-edge-removal top-3.

certified, not empirical: at $\varepsilon\leq0.1$ few nodes flip for either model, so the penalty buys a larger certified margin, not better small-budget accuracy. Attack and defense thus couple through one operator: across 10 seeds the attack magnitude and certified radius are anticorrelated (-0.65 ± 0.12, negative in $\mathbf{10/10}$, $p\approx10^{-160}$).

6 Case Study: Auditing a Fraud Detector

GNN fraud detectors are a prime adversarial target: a flagged account can slip past detection by perturbing a few behavioural-similarity edges (Zügner, Akbarnejad, and Günnemann 2018; Dou et al. 2020). AEGIS audits such a detector directly, with no labels and no retraining, reading off from a *single* S_c query which edges most threaten its predictions. Figure 6 shows the audit on an Amazon Fraud cluster: the edge-weighted ranking $A_{ij}v_{ij}$ (Proposition 3) pinpoints the fault lines, reproducing the brute-force single-edge-removal ranking (τ=1.0 here) at the cost of one query rather than $|E|$ reconvergences. The audit is *non-circular*: the ranking transfers across the full fraud graph (fig. 5, where uniform edge weights are least informative, so v_{ij} adds the most, $\Delta\tau$ up to $+\mathbf{0.65}$) and the SVD direction transfers from a separately trained surrogate (section 5.1), confirming v_{ij} is model-intrinsic, not a gradient artifact. Full walk-through in appendix G.

7 Related Work

Each of three threads supplies at most one of AEGIS's diagnostics. **Structural attacks**, among them Nettack (Zügner, Akbarnejad, and Günnemann 2018), Mettack (Zügner and Günnemann 2019a), GR-BCD/PR-BCD (Geisler et al. 2021) and variants (Wu et al. 2019; Xu et al. 2019a), optimise surrogate gradients to flip predictions but yield no analytic sensitive direction and no per-node radius; sober-look studies (Mujkanovic et al. 2022; Gosch et al. 2023) flag the evaluation pitfalls we adopt (appendix F.5).

Certified robustness and defense. Convex relaxations (Zügner and Günnemann 2019b), smoothing (Bojchevski, Klicpera, and Günnemann 2020; Bojchevski and Günnemann 2019; Schuchardt et al. 2023), collective (Schuchardt et al. 2021) and majority-vote (Li and Wang 2025) certificates, and weight-norm bounds (Abbahaddou et al. 2024) certify per-node robustness but do not name vulnerability-driving edges; empirical defenses (Zhu et al. 2019; Jin et al. 2020; Zhang and Zitnik 2020) harden models without surfacing per-edge sensitivity. Graph conformal prediction (Zargarbashi, Antonelli, and Bojchevski 2023) certifies a *fixed* graph, whereas AEGIS-Conformal covers the adversarial ε-ball via the S_c score-shift bound (section 4.1).

Implicit networks and influence functions. DEQs (Bai, Kolter, and Koltun 2019), IGNN (Gu et al. 2020), monotone equilibria (Winston and Kolter 2020), and well-posedness analyses (El Ghaoui et al. 2021) address *input* sensitivity $\partial z^*/\partial x$; AEGIS targets *structural* sensitivity $\partial Z/\partial A$, applying the Koh–Liang influence lineage (Koh and Liang 2017) and implicit differentiation (Lorraine, Vicol, and Duvenaud 2020; Gould, Hartley, and Campbell 2021) to a *structural-parameter* perturbation via P_c rather than to a loss. GNN explainers (Ying et al. 2019) rank edges for *fidelity*, not vulnerability; matrix perturbation theory (Stewart and Sun 1990; Trefethen and Embree 2005) supplies the pseudospectral toolkit; and curvature/over-squashing (Topping et al. 2022; Di Giovanni et al. 2023) and stability bounds (Gama, Bruna, and Ribeiro 2020; Kenlay, Thanou, and Dong 2021) tie edge structure to sensitivity only in aggregate, whereas AEGIS localises it per edge and node.

8 Conclusion

From one matrix-free object, S_c, AEGIS *audits*, *certifies*, and *defends* any GNN with continuous edge-weight message passing, adding the safety boundary of Theorem 1 for contractive implicit models. Three once-separate tools become readings of a single operator, coupled by construction.

Limitations. The radii r_v are first-order thresholds, and AEGIS-Conformal runs densely at N=200. Standard GAT and binary-mask architectures fall outside the framework (section 5.1), the $\varepsilon_{\text{crit}}$ track costs ∼6% accuracy, and insertion attacks are scoped out (section 2). AEGIS also audits vulnerability, not physics: a contractive surrogate cannot model voltage collapse, so it complements rather than replaces power-flow contingency screening (Donon et al. 2019; Nakiganda et al. 2023; Varbella, Gjorgiev, and Sansavini 2024).

Disclosure. We propose a 90-day coordinated-notification window: the diagnostic path (r_v, v_{ij}) releases unconditionally, while attack-direction reconstruction (algorithm 1) is gated behind ethics review, since it alone yields the damaging δA^* (Proposition 1).

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Appendix

Table of Contents

A Notation and Preliminaries

This appendix proves every result stated in the main paper and reports the supporting experiments in full. Its purpose is to let a reader reconstruct each claim end to end, so the proofs are written as explicit derivations rather than sketches. To keep them readable we fix all notation once here: every symbol used later appears in table 6, every abbreviation in the glossary below, and the standing assumptions and operator identities that recur across proofs are stated a single time and then invoked by name.

A.1 Symbols

The framework studies one object, the equilibrium of a graph operator, and how it moves when the graph is perturbed. Table 6 lists the symbols in the order a reader meets them: first the operator and its two Jacobians, then the sensitivity matrices read off them, then the two perturbation budgets and the constants that relate them, and finally the classification head and the conformal quantities.

Abbreviations. IFT, implicit function theorem; JVP/VJP, Jacobian–vector and vector–Jacobian products (matrix-free directional derivatives); SVD/rSVD, (randomized) singular value decomposition; PGD, projected gradient descent; BCD, block coordinate descent; MC, Monte Carlo; TPS/APS, threshold and adaptive prediction-set conformity scores. Norms are $\|\cdot\|_2$ (spectral, on operators; Euclidean, on vectors) and $\|\cdot\|_F$ (Frobenius); $\rho(\cdot)$ is the spectral radius and $\otimes$ the Kronecker product.

A.2 Standing assumptions

All proofs operate on the same operator and in the same certified regime, recorded once here.

Definition 1 (Operator and certified regime)**.** *Let $F_\theta(Z, \hat{A}) = \phi(\hat{A}ZW^\top + X_{\text{proj}})$ with equilibrium $z^* = F_\theta(z^*, \hat{A})$. We assume:*

- *(A1)* Lipschitz activation. *ϕ is 1-Lipschitz (ReLU or similar), so $\|\text{diag}(\phi')\|_2 \le 1$ pointwise; the nonsmoothness of ReLU at 0 is handled by the conservative calculus of Bolte and Pauwels (2021), which supplies an IFT on each linear region.*
- *(A2)* Bounded weight. *$\|W\|_2 \le c$ through spectral normalization. This caps the* numerator *of g_W but does not by itself bound $\rho(W)$ away from 0 (see appendix D), so (A2) does not imply (A3).*
- *(A3)* Trained contractivity. *$\kappa := \|J_z\|_2 < 1$, equivalently $\left\|\hat{A}\right\|_2 \|W\|_2 < 1$, verified post-training ($\kappa \in [0.14, 0.59]$ across our suite).*
- *(A4)* All-active operating point. *On the operating set every unit is active, $\phi' \equiv 1$, so $J_z = \hat{A} \otimes W$. Only the* upper *side of the boundary theorem (appendix D) invokes (A4); every certificate uses (A1)–(A3) alone.*

Unless stated otherwise we work in the certified regime $\varepsilon < \varepsilon_{\text{crit}}$, where the perturbed operator is still a contraction and the resolvent $(I - J_z)^{-1}$ is well defined.

A.3 The objects the theory reasons about

The operator F_θ maps a node-state matrix Z to a new one by one round of graph propagation and a weight W; its fixed point z^* is the equilibrium an implicit GNN settles into. Two Jacobians describe how that equilibrium responds to a perturbation. The *state* Jacobian $J_z = \text{diag}(\phi')(\hat{A} \otimes W)$ measures how the next state depends on the current one; under (A3) it is a contraction, so a perturbation injected into the loop decays geometrically. The *structural* Jacobian $J_A = \partial F / \partial \text{vec}(A)$ measures how the operator depends directly on the graph. Differentiating the fixed-point equation $z^* = F_\theta(z^*, \hat{A})$ through the IFT combines the two into the structural sensitivity

$$S = (I - J_z)^{-1} J_A, \tag{11}$$

the single object every diagnostic is read from. The inverse resolvent $(I - J_z)^{-1}$ is what turns a local Jacobian into the equilibrium response: it sums the effect of the perturbation as it circulates around the loop. Restricting the input to feasible (symmetric, edge-supported) perturbations through the projection P_c gives the constrained operator $S_c = (I - J_z)^{-1} J_A P_c$ used in practice.

From S and S_c the paper reads three diagnostics, each proved in a later section: the certified budget $\varepsilon_{\text{crit}}$ below which the prediction cannot break (section B and appendix D); the per-node radius r_v that turns the margin of node v into a safe perturbation distance (section B); and the per-edge score $v_k = \|[S_c]_{:,k}\|_2$ that ranks how much each edge matters (section C). The conformal certificate of section E upgrades the first-order radius into a distribution-free coverage guarantee.

A.4 Matrix-free evaluation of the sensitivity operator

None of the diagnostics forms S explicitly, which is what lets the pipeline scale; the routing is algorithm 1 and we record here why a single matrix–vector rule suffices. Applying S_c to a vector v expands the resolvent as a Neumann series,

$$S_c\, v = (I - J_z)^{-1} J_A P_c\, v = \sum_{j=0}^{\infty} J_z^j\, J_A P_c\, v, \tag{12}$$

which converges because $\kappa < 1$ (A3). Truncating at depth K leaves a tail bounded by the same contraction factor,

$$\left\| S_c v - \sum_{j=0}^{K} J_z^j J_A P_c v \right\|_2 \le \frac{\kappa^{K+1}}{1-\kappa} \|J_A P_c v\|_2, \tag{13}$$

so the depth needed for a target accuracy grows only logarithmically in $1/\kappa$ ($K \in [20, 50]$ for $\kappa < 0.8$). Each term $J_z^j(\cdot)$ is one forward-mode JVP costing $O(Nd)$ memory rather than the $O((Nd)^2)$ of a dense S, and a randomized SVD returns the leading pair (σ_1, v_1) that the attack and defense use. The well-separated singular spectrum (gap 0.39–0.50) is why one rSVD matches an iterative attacker; the empirical truncation residual $\kappa^{200} \in [10^{-105}, 10^{-48}]$ on the suite confirms (13) is slack at the depths we run.

Table 6: Notation used throughout the appendix. “First used” points to the result that introduces or relies on the symbol.

Symbol	Meaning	First used
Operator and Jacobians		
$\hat{A}$	symmetric normalized adjacency $D^{-1/2}(A+I)D^{-1/2}$	Theorem 1
W	tied weight matrix of the layer	Theorem 1
$\phi,\ \mathrm{diag}(\phi')$	1-Lipschitz activation; its derivative mask	(A1)
z^*	equilibrium (fixed point) of F_θ	Theorem 1
J_z	state Jacobian $\mathrm{diag}(\phi')(\hat{A}\otimes W)$	Theorem 1
J_A	structural Jacobian $\partial F/\partial \mathrm{vec}(A)$	Theorem 1
κ	contraction factor $\|J_z\|_2 < 1$	(A3)
Sensitivity read off the operator		
S	structural sensitivity $(I - J_z)^{-1}J_A$	Theorem 1
S_c	edge-constrained sensitivity $(I - J_z)^{-1}J_A P_c$	section 3
$P_c,\ P_E$	projection onto symmetric, edge-supported perturbations	Theorem 2
$S_v,\ S_{c,v}$	block rows of $S,\ S_c$ at node v	Proposition 2
$\sigma_1(\cdot)$	leading singular value	Proposition 1
$w_k,\ v_k$	edge weight $[\hat{A}]_{ij}$; edge score $\|[S_c]_{:,k}\|_2$	Proposition 3
d_k	true damage of removing edge k	Proposition 3
Budgets and the constants that relate them		
$\varepsilon_{\mathrm{crit}}$	norm certificate $(1-\kappa)/\|W\|_2$	Theorem 1
$\varepsilon_{\mathrm{spec}}$	all-active spectral break $1/\rho(W) - \rho(\hat{A})$	Theorem 2
$\varepsilon_{\mathrm{br}},\ \varepsilon_{\mathrm{br}}^{\mathrm{all}}$	deployed / all-active contraction-break budget	Theorem 2
$\varepsilon_{\mathrm{reach}}$	measured nonlinear break budget	Remark 2
g_W	nonnormality of W, $\|W\|_2/\rho(W) \geq 1$	Theorem 2
η	Neumann-to-spectral slack (pseudospectral index)	Observation 1
$\beta = \sigma_E$	edge-supported mass of the Perron mode u_1	Theorem 2
C	bracket constant $g_W(1+\kappa)/(1-\kappa)$	Theorem 2
L_J	equilibrium-map curvature, $\leq \|W\|_2^2\,\|z^*\|$	Proposition 3
Classification head and conformal prediction		
$f = Wz_v^* + b$	node logits read by the linear head	Proposition 2
$m_v^{(c)}$	logit margin $f_{y_v} - f_c$ to competitor c	Proposition 2
r_v	per-node first-order sensitivity radius	Proposition 2
$\mathrm{score}^{\mathrm{TPS}},\ \mathrm{score}^{\mathrm{APS}}$	the two conformity scores	Definition 2
$C(v),\ \hat{q}$	prediction set; conformal threshold	Definition 2
$L_1^{(c)},\ C_v$	first-order and curvature score-shift constants	Lemma 1
$\alpha,\ n_{\mathrm{cal}}$	target miscoverage; calibration-set size	Theorem 4

B Equilibrium Sensitivity and the Phase Transition

This section proves the three results that turn the operator of section A into a safety boundary: how the equilibrium moves under a perturbation, how far the worst-case slack between the certificate and the true threshold can grow, and the per-node radius that the certificate yields. Each proof restates its claim in brief and then derives it step by step using the notation of table 6.

B.1 Proof of the Phase Transition

Theorem 1 states that, under (A1)–(A3), a structural perturbation of size $\varepsilon = \left\|\delta\hat{A}\right\|_F$ passes through three regimes set by the critical budget $\varepsilon_{\mathrm{crit}} = (1-\kappa)/\|W\|_2$: subcritical ($\varepsilon < \varepsilon_{\mathrm{crit}}$, the equilibrium moves boundedly), critical ($\varepsilon \to \varepsilon_{\mathrm{crit}}$, its sensitivity diverges), and supercritical ($\varepsilon \geq \varepsilon_{\mathrm{crit}}$, contraction is no longer certified).

Proof. Step 1: the subcritical sensitivity. On each ReLU linear region F_θ is affine, and the activation pattern at z^* is stable for small $\left\|\delta\hat{A}\right\|$ because the set of adjacencies whose equilibrium lands exactly on a region face has Lebesgue measure zero. The conservative calculus of Bolte and Pauwels (2021) therefore supplies an IFT on each region. Differentiating the fixed-point condition $G(z,\hat{A}) = z - F(z,\hat{A}) = 0$ at $(z^*,\hat{A})$ gives the equilibrium response

$$\Delta z^* = (I - J_z)^{-1}J_A\,\mathrm{vec}(\delta\hat{A}) + O(\left\|\delta\hat{A}\right\|^2), \tag{14}$$

which is exactly $S\,\mathrm{vec}(\delta\hat{A})$ to first order. The resolvent factor is what makes this an equilibrium rather than a one-step response: a perturbation injected into the loop returns attenuated by J_z on each pass, and the total is the geometric sum $\sum_{j\geq 0} J_z^j = (I - J_z)^{-1}$. Since $\kappa = \|J_z\|_2 < 1$ this sum converges with

$$\left\|(I - J_z)^{-1}\right\|_2 \leq \sum_{j\geq 0}\kappa^j = \frac{1}{1-\kappa}, \tag{15}$$

so taking the leading singular value in (14) yields the shift bound $\|\Delta z^*\|_F \le \sigma_1(S)\,\varepsilon + O(\varepsilon^2)$ with $\sigma_1(S) \le \|J_A\|_2/(1-\kappa)$, i.e. (5).

Step 2: contraction survives below $\varepsilon_{\text{crit}}$. Perturbing $\hat{A}$ by $\delta\hat{A}$ changes the state Jacobian to $J'_z = \operatorname{diag}(\phi')(\hat{A}' \otimes W)$. Because $\|\operatorname{diag}(\phi')\|_2 \le 1$ (A1) and the spectral norm is Kronecker-multiplicative,

$$\|J'_z\|_2 \le \left\|\hat{A}'\right\|_2 \|W\|_2 \le (\left\|\hat{A}\right\|_2 + \varepsilon)\,\|W\|_2\,, \qquad (16)$$

and the right-hand side is below 1 precisely when $\varepsilon < 1/\|W\|_2 - \left\|\hat{A}\right\|_2 = \varepsilon_{\text{crit}}$. A spectral-norm contraction has a unique fixed point by the Banach theorem, so for every feasible $\varepsilon < \varepsilon_{\text{crit}}$ the perturbed model still has a single well-defined equilibrium: this is the subcritical regime, and $\varepsilon_{\text{crit}}$ is a sufficient safety boundary.

Step 3: the critical divergence and why $\varepsilon_{\text{crit}}$ is one-sided. For any matrix M the resolvent $(I-M)^{-1}$ has eigenvalues $(1-\lambda_i(M))^{-1}$, so

$$\left\|(I-M)^{-1}\right\|_2 \;\ge\; \rho\big((I-M)^{-1}\big) = \frac{1}{\min_i |1-\lambda_i(M)|}, \qquad (17)$$

which blows up when an eigenvalue of M approaches 1. The rate depends on the geometry of J'_z. When J'_z is normal with a dominant real-positive (Perron) eigenvalue equal to $\|J'_z\|_2$, the case for symmetric $\hat{A}$ and W, the spectral gap to 1 is $\min_i |1-\lambda_i| = 1 - \|J'_z\|_2 = \|W\|_2(\varepsilon_{\text{crit}} - \varepsilon)$ by (16) with equality, giving the $\Omega\big(1/(\varepsilon_{\text{crit}}-\varepsilon)\big)$ divergence. The certificate is only one-sided because $\|J'_z\|_2 \to 1$ need not drive the gap to zero: $M = \operatorname{diag}(-s, 0)$ has $\|M\|_2 = s$ yet $\left\|(I-M)^{-1}\right\|_2 = 1$, a dominant eigenvalue at $-s$ that stays far from 1. So $\varepsilon_{\text{crit}}$ lower-bounds the true divergence threshold, and the slack is governed by how far J'_z is from normal, the nonnormality η of Observation 1.

Step 4: supercritical. For $\varepsilon \ge \varepsilon_{\text{crit}}$ the bound (16) no longer forces $\|J'_z\|_2 < 1$, the Banach contraction certificate fails, and the iteration may converge to a different fixed point, oscillate, or diverge. □

B.2 A graph-independent bound on the nonnormality

The slack η in Step 3 could in principle grow with the graph; the next observation shows it does not, in the all-active case it is set by W alone.

Observation 1 (Graph-independent nonnormality bound, all-active case)**.** *For $J_z = \operatorname{diag}(\phi')(\hat{A} \otimes W)$ with $\hat{A}$ symmetric and all units active ($\phi' \equiv 1$), the nonnormality obeys $\eta \le \kappa(V_W)$, where $W = V_W D_W V_W^{-1}$ and $\kappa(V_W) = \|V_W\|_2 \left\|V_W^{-1}\right\|_2$ is the eigenvector conditioning of W, independent of $\hat{A}$.*

Proof. Under (A4), $J_z = \hat{A} \otimes W$, whose eigenvalues are the products $\lambda_i(\hat{A})\lambda_j(W)$ and whose eigenvectors are the Kronecker products of those of $\hat{A}$ and W (Stewart and Sun 1990). Symmetric $\hat{A}$ diagonalizes in an orthonormal basis, so its eigenvector matrix has conditioning $\kappa(U_{\hat{A}}) = 1$ and contributes nothing to the joint conditioning; the joint eigenvector matrix therefore has condition number $\kappa(V_W)$. For any diagonalizable $M = VDV^{-1}$ the resolvent obeys $\left\|(I-M)^{-1}\right\|_2 \le \kappa(V)/(1-\rho(M))$, and the slack relative to the spectral-radius rate $1/(1-\rho(M))$ is exactly $\kappa(V)$. Applied to J_z this gives $\eta \le \kappa(V_W)$. □

Remark 1 (Empirical extension to general ReLU patterns)**.** *The bound assumes $\phi' \equiv 1$. For general ReLU patterns the diagonalization argument no longer applies exactly, but the measured nonnormality stays moderate, $\eta \in [1.19, 2.47]$ across the suite (appendix F.5), tracking $\kappa(V_W)$ closely. Graph structure does not amplify it; the modest η traces to the spectral-norm regularization of W, exactly the quantity (A2) controls.*

B.3 Proof of the Per-Node Radius

Proposition 2 gives, for a node v classified as y_v with positive margins $m_v^{(c)} = f_{y_v} - f_c$ to every competitor c, the certified radius $r_v = \min_{c \ne y_v} m_v^{(c)} / \|(W_{y_v} - W_c)S_v\|_2$, below which the prediction cannot change.

Proof. A perturbation of size ε moves the node's equilibrium by $\Delta z_v^* \approx S_v \operatorname{vec}(\delta\hat{A})$ (Step 1 of section B.1). Because the head is linear, each margin moves by $(W_{y_v} - W_c)\Delta z_v^*$, and Cauchy–Schwarz bounds this by the operator norm of the composed map times the perturbation size,

$$\begin{aligned}\left|\Delta m_v^{(c)}\right| &= \left|(W_{y_v} - W_c)\,S_v \operatorname{vec}(\delta\hat{A})\right| \\ &\le \|(W_{y_v} - W_c)S_v\|_2 \left\|\delta\hat{A}\right\|_F. \qquad (18)\end{aligned}$$

The prediction of v survives as long as every margin stays positive. The margin to competitor c first reaches zero at $\left\|\delta\hat{A}\right\|_F = m_v^{(c)} / \|(W_{y_v} - W_c)S_v\|_2$, and taking the binding (smallest) value over competitors gives

$$r_v = \min_{c \ne y_v} \frac{m_v^{(c)}}{\|(W_{y_v} - W_c)S_v\|_2}. \qquad (19)$$

The form is a margin divided by how fast the worst perturbation spends it: the denominator is the product of two amplifiers, the decision-boundary gap $W_{y_v} - W_c$ and the equilibrium sensitivity S_v, so a node earns a large radius exactly when it has a wide margin and a low structural sensitivity. Minimizing over all competitors measures the distance to the nearest first-order decision boundary, so the guarantee holds even if the runner-up class changes along the way; substituting the constrained $S_{c,v}$ for S_v tightens the denominator and yields the larger constrained radius. □

C From Continuous Sensitivity to Discrete Rankings

The sensitivity S_c is a continuous, implicit-model object, while practitioners delete whole edges and use explicit, finite-depth GNNs. Two bridges carry S_c across that gap, and we prove both here: the first shows the continuous score ranks discrete edge removals, the second shows the same construction applies to unrolled explicit networks.

C.1 Continuous-to-Discrete Ranking Transfer

Proposition 3 states that, when every single-edge removal is subcritical, the true removal damage of edge k factors as $d_k = w_k v_k + R_k$ with $|R_k| \le L_J w_k^2 / \big(2(1-\kappa)^2\big)$, so the edge-weighted score $w_k v_k$ preserves the brute-force removal order whenever the score gap between two edges beats this quadratic remainder. The pairwise condition under which a pair (k_1, k_2) with $v_{k_1} > v_{k_2}$ and $w_{k_1} \ge w_{k_2}$ keeps its order is

$$w_{k_1} < \frac{v_{k_1} - v_{k_2}}{L_J/(1-\kappa)^2}. \tag{20}$$

Proof. Step 1: a removal is a continuous edge-weight perturbation. Deleting edge k under fixed normalization sets $[\delta\hat{A}]_{ij} = [\delta\hat{A}]_{ji} = -w_k$ with the degree matrix D held fixed, which is exactly the continuous perturbation S_c is built to score. By the first-order sensitivity (14) restricted to the edge, the damage is $\|\Delta z^*\| \approx w_k v_k$ with $v_k = \|[S_c]_{:,k}\|_2$. (Recomputing the normalization instead adds an $O(d_i^{-1})$ rescaling of the incident edges, negligible away from low-degree nodes.) It remains to bound the second-order error R_k.

Step 2: the remainder is governed by one cross term, with nothing omitted. On the interior of a ReLU linear region the activation mask $D = \mathrm{diag}(\phi')$ is constant, so the operator $F(z, \hat{A}) = D\,(\hat{A} \otimes W)\,z + Dx$ is *bilinear* in $(\hat{A}, z)$: linear in z at fixed $\hat{A}$, and linear in $\hat{A}$ at fixed z. Both pure second partials therefore vanish,

$$F_{zz} = \frac{\partial^2 F}{\partial z^2} = 0, \qquad F_{AA} = \frac{\partial^2 F}{\partial \mathrm{vec}(A)^2} = 0, \tag{21}$$

and the only surviving second derivative is the adjacency–state cross term F_{zA}. Differentiating the IFT sensitivity $\partial z^*/\partial \mathrm{vec}(A) = (I - J_z)^{-1} J_A$ once more and using (21) to drop every other term gives the full Hessian of the equilibrium map,

$$\frac{\partial^2 z^*}{\partial \mathrm{vec}(A)^2} = (I - J_z)^{-1}\big(F_{zA} + F_{Az}\big)(I - J_z)^{-1} J_A. \tag{22}$$

Each summand carries two resolvents $(I - J_z)^{-1}$, a cross partial F_{zA} or F_{Az} proportional to $\|W\|_2$, and J_A proportional to $\|W\|_2\,\|z^*\|$ (since $J_A = \partial F/\partial \mathrm{vec}(A) \propto z^* W^\top$). Bounding the resolvents by $1/(1-\kappa)$ each (eq. (15)) and collecting the weight factors into

$$L_J \le \|W\|_2^2\,\|z^*\|, \qquad \|z^*\| \le \frac{\|X_{\mathrm{proj}}\|}{1-\kappa}, \tag{23}$$

where the bound on $\|z^*\|$ is the κ-contraction of F at its fixed point (A3), yields $\big\|\partial^2 z^*/\partial \mathrm{vec}(A)^2\big\|_2 \le (1-\kappa)^{-2} L_J$. The feasible segment $\hat{A} \to \hat{A} + \delta\hat{A}$ crosses finitely many regions (simultaneous boundary crossings are non-generic, measure zero), and the conservative IFT of Bolte and Pauwels (2021) applies on each, so summing the second-order Taylor remainder across them preserves the bound:

$$|R_k| \le \tfrac{1}{2}\,(1-\kappa)^{-2} L_J\, w_k^2. \tag{24}$$

This is the *complete* second-order remainder: the bilinearity (21) leaves no other second-derivative term to bound, so (24) is exact on each region rather than a dominant-term reduction.

Step 3: the score preserves the ranking. With $d_k = w_k v_k + R_k$ from Steps 1–2 and $C := L_J/\big(2(1-\kappa)^2\big)$, two edges differ by

$$d_{k_1} - d_{k_2} \;\ge\; \big(w_{k_1} v_{k_1} - w_{k_2} v_{k_2}\big) - C\big(w_{k_1}^2 + w_{k_2}^2\big). \tag{25}$$

When $v_{k_1} > v_{k_2}$ and $w_{k_1} \ge w_{k_2}$ the first-order gap is positive, and condition (20) is exactly what makes it exceed the summed quadratic remainders, so the sign of $d_{k_1} - d_{k_2}$ matches that of the score gap and the order is preserved. Because the signal scales as $O(w_k)$ and the error only as $O(w_k^2)$, the ranking is reliable for the small-weight edges of sparse graphs; empirically $|R_k|$ runs 2–10× below $L_J w_k^2$, and the edge-weighted score reaches Kendall $\tau = +0.996$ against brute-force single-edge removal on full-graph Amazon Photo (section 5.1). ☐

C.2 Structural Sensitivity for K-Layer Explicit GNNs

Proposition 4 states that an explicit K-layer network $Z_K = g_K \circ \cdots \circ g_1(X, A)$, which has no fixed point, still admits the unrolled sensitivity S_K of eq. (10), and that S_K replaces S throughout the pipeline.

Proof. Step 1: unroll and apply the chain rule. Write Z_K as the composition of the K layers and differentiate $\mathrm{vec}(Z_K)$ with respect to $\mathrm{vec}(A)$. Each layer l depends on A both directly, through $J_A^{(l)} = \partial g_l/\partial \mathrm{vec}(A)$, and indirectly through its input Z_{l-1}; propagating the direct dependence of layer l forward through the later layers' input Jacobians $J_z^{(k)} = \partial g_k/\partial Z_{k-1}$ and summing over l gives

$$S_K = \frac{\partial\,\mathrm{vec}(Z_K)}{\partial\,\mathrm{vec}(A)} = \sum_{l=1}^{K} \Big(\prod_{k=l+1}^{K} J_z^{(k)}\Big) J_A^{(l)}, \tag{26}$$

which is (10).

Step 2: the first-order shift bound. Submultiplicativity of the spectral norm applied term by term in (26) gives $\sigma_1(S_K) \le \sum_{l=1}^{K} \big(\prod_{k=l+1}^{K} \big\|J_z^{(k)}\big\|_2\big) \big\|J_A^{(l)}\big\|_2$, and a first-order Taylor expansion of Z_K in A turns this into the shift bound $\|\Delta Z_K\|_F \le \sigma_1(S_K)\,\|\delta A\|_F + O(\|\delta A\|^2)$.

Step 3: the implicit limit. For a weight-tied model every $\big\|J_z^{(k)}\big\|_2 \le \kappa$, so the sum is geometric,

$$\sigma_1(S_K) \le \|J_A\|_2 \sum_{j=0}^{K-1} \kappa^j = \|J_A\|_2\,\frac{1-\kappa^K}{1-\kappa} \xrightarrow[K\to\infty]{} \frac{\|J_A\|_2}{1-\kappa}, \tag{27}$$

recovering the implicit bound of section B.1: unrolling a finite network is the truncated version of the geometric series that the implicit resolvent $(I - J_z)^{-1}$ sums in closed form, so the two agree as depth grows. Because S_K enters the constrained construction S_c and the per-node radius r_v (Proposition 2) in exactly the same place as S, the whole pipeline carries over; only the closed-form $\varepsilon_{\mathrm{crit}}$, which needs the contraction (A3), stays restricted to the implicit subclass. Empirically the explicit models match the implicit tightness (0.99–1.02 across the suite, appendix F.6). □

D The Two-Sided Contraction Boundary

Section B.1 certified safety *below* $\varepsilon_{\mathrm{crit}}$. This section bounds the contraction boundary from *above* as well, proving the full version of Theorem 2: the budget at which the operator first loses contraction sits between $\varepsilon_{\mathrm{crit}}$ and a constant multiple of it. The lower side holds for any 1-Lipschitz ϕ and is the certificate the paper sells; the upper side is proved for the all-active operator (A4), and the gap between them is a single computable constant. We work under Definition 1, with $\hat{A}$ symmetric and $\varepsilon_{\mathrm{crit}} > 0$.

D.1 Two budgets and the gap between them

Two budgets bound the boundary from opposite sides, the *norm certificate* $\varepsilon_{\mathrm{crit}}$ and the *all-active spectral break* $\varepsilon_{\mathrm{spec}}$:

$$\varepsilon_{\mathrm{crit}} = \frac{1-\kappa}{\|W\|_2} = \frac{1}{\|W\|_2} - \left\|\hat{A}\right\|_2, \tag{28}$$

$$\varepsilon_{\mathrm{spec}} = \frac{1}{\rho(W)} - \rho(\hat{A}). \tag{29}$$

The certificate $\varepsilon_{\mathrm{crit}}$ uses the operator norm $\|W\|_2$, which is the worst case over all directions, so it is conservative; the spectral budget $\varepsilon_{\mathrm{spec}}$ uses the spectral radius $\rho(W)$, which is the actual unstable direction, so it is exact for the all-active operator. The whole gap between them is supplied by the nonnormality of W,

$$\varepsilon_{\mathrm{spec}} - \varepsilon_{\mathrm{crit}} = \left(\tfrac{1}{\rho(W)} - \tfrac{1}{\|W\|_2}\right) + \left(\left\|\hat{A}\right\|_2 - \rho(\hat{A})\right) \;\geq\; 0, \tag{30}$$

where the second bracket vanishes because $\hat{A}$ is symmetric, leaving only the term driven by $g_W := \|W\|_2/\rho(W) \geq 1$.

The quantity g_W is audited per model rather than controlled a priori: spectral normalization (A2) caps $\|W\|_2$ but not $\rho(W)$, so in principle $\rho(W) \to 0$ at fixed $\|W\|_2$ would send $g_W \to \infty$. We therefore measure g_W on each trained model ($g_W \in [1.19, 2.47]$ on the suite, with the a-posteriori certificate $g_W \leq \kappa_2(V_W)$, the eigenvector conditioning of W) and report the bracket constant as a function of that value. Every constant below is computable post-training from $\rho(W)$, $\|W\|_2$, $\left\|\hat{A}\right\|_2$ and the edge set.

D.2 Edge-feasible critical direction

A break must be driven by a feasible perturbation, one that is symmetric and supported on existing edges. Let P_E be the orthogonal projection onto that subspace $\mathcal{S}_E = \{M = M^\top : M_{ij} = 0 \text{ if } (i,j) \notin E \cup \{(i,i)\}\}$, let u_1 be a unit top eigenvector of $\hat{A}$, and set $g_E := P_E(u_1 u_1^\top)$, $\sigma_E := \|g_E\|_F$, and the unit feasible direction $B := g_E/\sigma_E$. The alignment of this feasible direction with the critical rank-one mode is

$$\beta := \langle u_1, B u_1 \rangle = \frac{\langle u_1, P_E(u_1 u_1^\top) u_1 \rangle}{\sigma_E} = \frac{\|g_E\|_F^2}{\sigma_E} = \sigma_E, \tag{31}$$

using that P_E is an orthogonal projection and $u_1 u_1^\top$ is symmetric. So the alignment β equals the edge-supported Frobenius mass σ_E of the critical direction, which is strictly positive whenever the Perron mode places mass on the edge set, true for every connected graph ($u_1 > 0$ entrywise, $E \neq \emptyset$). The hypothesis $\beta > 0$ is therefore an observable, not an assumption.

D.3 The bracket theorem

Theorem 3 (Constant-factor two-sided bracket for the all-active IGNN contraction boundary, full statement)**.** *Let Definition 1 hold with $\varepsilon_{\mathrm{crit}} > 0$ and $\hat{A}$ symmetric, and let $\beta = \sigma_E \in (0,1]$ be the edge-supported mass of the critical mode. Then:*

(i) ***Lower side (any 1-Lipschitz ϕ; no (A4)).*** *For every feasible $\delta\hat{A}$ with $\left\|\delta\hat{A}\right\|_F \leq \varepsilon < \varepsilon_{\mathrm{crit}}$, the perturbed operator is an $\|\cdot\|_2$-contraction with a unique equilibrium and $\rho(J_z') < 1$. Hence $\varepsilon_{\mathrm{br}} \geq \varepsilon_{\mathrm{crit}}$, and in particular $\varepsilon_{\mathrm{br}}^{\mathrm{all}} \geq \varepsilon_{\mathrm{crit}}$.*

(ii) ***Upper side (all-active, requires (A4)).*** *The feasible rank-one $\delta\hat{A}^\star = (\varepsilon_{\mathrm{spec}}/\beta)\, B \in \mathcal{S}_E$, of size $\left\|\delta\hat{A}^\star\right\|_F = \varepsilon_{\mathrm{spec}}/\beta$, drives the all-active spectral radius to 1, so $\varepsilon_{\mathrm{br}}^{\mathrm{all}} \leq \varepsilon_{\mathrm{spec}}/\beta$. In the unconstrained-symmetric threat model ($P_E = I$, $\beta = 1$) it is exact: $\varepsilon_{\mathrm{br}}^{\mathrm{all}} = \varepsilon_{\mathrm{spec}}$ with extremizer $\varepsilon_{\mathrm{spec}}\, u_1 u_1^\top$.*

(iii) ***Constant-factor enclosure.*** *$\varepsilon_{\mathrm{spec}} \leq C\, \varepsilon_{\mathrm{crit}}$ with the explicit, dimension-free, a-posteriori-computable constant*

$$C := g_W\, \frac{1+\kappa}{1-\kappa} = \frac{\|W\|_2}{\rho(W)} \cdot \frac{1 + \left\|\hat{A}\right\|_2 \|W\|_2}{1 - \left\|\hat{A}\right\|_2 \|W\|_2},$$

so the all-active boundary obeys

$$\boxed{\varepsilon_{\mathrm{crit}} \;\leq\; \varepsilon_{\mathrm{br}}^{\mathrm{all}} \;\leq\; \tfrac{C}{\beta}\, \varepsilon_{\mathrm{crit}}}. \tag{32}$$

The bracket is non-vacuous whenever $\beta > 0$; on the suite $\beta \approx 0.62$, giving $C/\beta \lesssim 16$. The β-free form $\varepsilon_{\mathrm{crit}} \leq \varepsilon_{\mathrm{br}}^{\mathrm{all}} \leq C\, \varepsilon_{\mathrm{crit}}$ holds exactly for the unconstrained-symmetric model ($\beta = 1$).

(iv) ***Equality conditions.*** *(a) the endpoints coincide, $\varepsilon_{\mathrm{crit}} = \varepsilon_{\mathrm{spec}}$, iff $g_W = 1$ (W normal), independent of κ, β; (b) all three coincide, $\varepsilon_{\mathrm{crit}} = \varepsilon_{\mathrm{br}}^{\mathrm{all}} = \varepsilon_{\mathrm{spec}}$, iff $g_W = 1$ and $\beta = 1$; (c) the slack constant is one, $C/\beta = 1$, iff additionally $\kappa \to 0^+$.*

Proof. Setup. Feasible perturbations lie in $\mathcal{S}_E$ with $\left\|\delta\hat{A}\right\|_F = \varepsilon$. We use Weyl's inequality, $\rho(\hat{A}') \leq \rho(\hat{A}) + \left\|\delta\hat{A}\right\|_2$ and $\left\|\hat{A}'\right\|_2 \leq \left\|\hat{A}\right\|_2 + \left\|\delta\hat{A}\right\|_2$, together with $\left\|\delta\hat{A}\right\|_2 \leq \left\|\delta\hat{A}\right\|_F = \varepsilon$. Since $\hat{A}$ and the extremal $\delta\hat{A}$ are symmetric, $\rho(\hat{A}') = \left\|\hat{A}'\right\|_2$ and Weyl holds with equality along an aligned rank-one mode.

(i) Lower side. By (A1), $\|\mathrm{diag}(\phi')\|_2 \le 1$; the spectral norm is sub-multiplicative and Kronecker-multiplicative, so as in (16), $\|J_z'\|_2 \le (\left\|\hat{A}\right\|_2 + \varepsilon)\,\|W\|_2$, which is below 1 for $\varepsilon < \varepsilon_{\mathrm{crit}}$. A spectral-norm contraction has a unique fixed point and $\rho(J_z') \le \|J_z'\|_2 < 1$, so no feasible $\varepsilon < \varepsilon_{\mathrm{crit}}$ breaks contraction of the deployed model: $\varepsilon_{\mathrm{br}} \ge \varepsilon_{\mathrm{crit}}$. This step uses (A1)–(A3) only. The additive gap (30) then gives $\varepsilon_{\mathrm{crit}} \le \varepsilon_{\mathrm{spec}}$.

(ii) Upper side, all-active. Let u_1 be a unit top eigenvector of symmetric $\hat{A}$. Along the rank-one symmetric direction $t\,u_1u_1^\top$ ($\left\|u_1u_1^\top\right\|_F = 1$), $\rho(\hat{A} + t\,u_1u_1^\top) = \rho(\hat{A}) + t$ exactly, so under (A4), $\rho(J_z') = (\rho(\hat{A}) + t)\rho(W)$ reaches 1 at $t = \varepsilon_{\mathrm{spec}}$, giving the unconstrained break budget $\varepsilon_{\mathrm{spec}}$ with extremizer $\varepsilon_{\mathrm{spec}}\,u_1u_1^\top$. For edge feasibility, project to $\mathcal{S}_E$ via $B = P_E(u_1u_1^\top)/\sigma_E$. The map $t \mapsto \lambda_{\max}(\hat{A}+tB) = \max_{\|v\|=1} v^\top(\hat{A} + tB)v$ is a pointwise maximum of affine functions, hence convex, so it lies above its tangent at $t = 0$, $\rho(\hat{A} + tB) \ge \rho(\hat{A}) + \beta t$ for all $t \ge 0$ (this lower direction is what convexity buys; a concave version fails on every random instance). Choosing $\delta\hat{A}^\star = (\varepsilon_{\mathrm{spec}}/\beta)B$ drives $\rho(\hat{A}') \ge \rho(\hat{A}) + \varepsilon_{\mathrm{spec}} = 1/\rho(W)$, so $\rho(J_z') \ge 1$ and $\varepsilon_{\mathrm{br}}^{\mathrm{all}} \le \left\|\delta\hat{A}^\star\right\|_F = \varepsilon_{\mathrm{spec}}/\beta$, exact when $\beta = 1$.

(iii) Closing the bracket. Dropping $-\rho(\hat{A}) \le 0$ and using $\rho(\hat{A}) = \left\|\hat{A}\right\|_2$, $\varepsilon_{\mathrm{spec}} \le 1/\rho(W) = g_W/\|W\|_2 = g_W(\varepsilon_{\mathrm{crit}} + \left\|\hat{A}\right\|_2)$. Since $\left\|\hat{A}\right\|_2 = \kappa/\|W\|_2 = \kappa(\varepsilon_{\mathrm{crit}} + \left\|\hat{A}\right\|_2)$ gives $\left\|\hat{A}\right\|_2 = \frac{\kappa}{1-\kappa}\varepsilon_{\mathrm{crit}}$, we obtain $\varepsilon_{\mathrm{spec}} \le g_W\frac{1}{1-\kappa}\varepsilon_{\mathrm{crit}} \le g_W\frac{1+\kappa}{1-\kappa}\varepsilon_{\mathrm{crit}} = C\,\varepsilon_{\mathrm{crit}}$. With (i) and (ii) the boxed bracket (32) follows.

(iv) Equality conditions. (a) The gap $\frac{1}{\rho(W)} - \frac{1}{\|W\|_2} \ge 0$ is zero iff $g_W = 1$, with κ, β absent. (b) Adding edge-feasibility of the exact extremizer $\varepsilon_{\mathrm{spec}}\,u_1u_1^\top$ requires $\beta = 1$, giving $\varepsilon_{\mathrm{crit}} = \varepsilon_{\mathrm{br}}^{\mathrm{all}} = \varepsilon_{\mathrm{spec}}$. (c) Forcing $\frac{1+\kappa}{1-\kappa} = 1$ in addition requires $\kappa \to 0^+$. □

This recovers and sharpens Theorem 1: the old result is the lower side (i), part (ii) supplies the matching all-active upper side, and (iv) separates the three equality conditions. The characterization is constant-factor (dimension-free), rate-sharp in the normal, edge-aligned corner (iv)(b), with enclosure constant C/β reaching 10–16× on the suite (random-instance $\varepsilon_{\mathrm{spec}}/\varepsilon_{\mathrm{crit}} \in [1.02, 18.2]$, median 2.25). Two identities, both numerically exact over thousands of random certified-regime instances, underlie it: the all-active spectral law $\rho(J_z') = \rho(\hat{A}')\rho(W)$ (A4 only), and the additive gap (30), whose slack is supplied entirely by W's non-normality when $\hat{A}$ is symmetric.

D.4 First-order tightness across the budget

The bracket rests on the first-order term being an accurate proxy for the true shift at the budgets we use; fig. 7 confirms this directly. The ratio of the true equilibrium shift to its first-order prediction stays within the operational budget and grows only as ε leaves the linear regime, which is exactly why the realistic-budget analysis below relies on the first-order $\sigma_1(S_c)$ rather than on proximity to criticality.

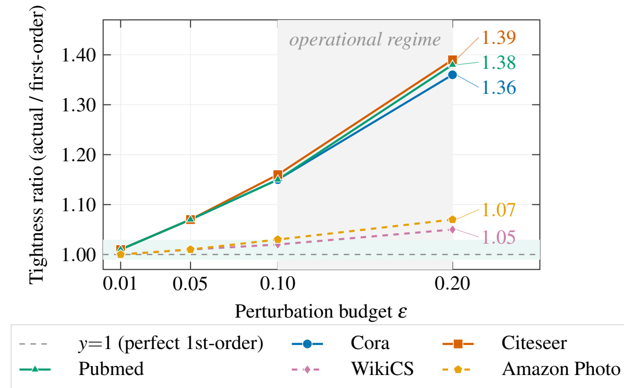

Figure 7: First-order tightness of the equilibrium-shift prediction versus the perturbation budget ε (full graph, mean over 10 seeds). The ratio is the true shift divided by the first-order prediction $\sigma_1(S)\varepsilon$; it is 1 when the linearization is exact and rises as ε grows. Within the operational regime ($\varepsilon \le 0.10$) every curve stays below 1.16; only at $\varepsilon = 0.20$ do citation graphs (solid) reach 1.36–1.39, while product graphs (dashed) stay within 1.07. This shift-prediction ratio is distinct from the ranking tightness quoted elsewhere (1.00–1.02, appendices F.5 and F.6), which is measured at the operational subgraph scale; the figure traces the budget dependence that motivates capping ε in that regime.

D.5 Nonlinear break budget: an empirical extension

Remark 2 (Observation O1: nonlinear break budget and the active fraction; empirical, not part of Theorem 2)**.** *The deployed model is ReLU, not all-active, and its measured nonlinear break budget* $\varepsilon_{\mathrm{reach}}$ *exceeds the all-active* $\varepsilon_{\mathrm{spec}}$*. Over the 10 preferred seeds (50-node ego subgraph,* $\left\|\hat{A}\right\|_F = 1.77$*): at* $\kappa_0 = 0.5$*,* $\varepsilon_{\mathrm{reach}}/\varepsilon_{\mathrm{spec}} = 1.41 \pm 0.12$ *and* $\varepsilon_{\mathrm{reach}}/\varepsilon_{\mathrm{crit}} = 2.17 \pm 0.18$*; at* $\kappa_0 = 0.9$*,* $\varepsilon_{\mathrm{reach}}/\varepsilon_{\mathrm{spec}} = 1.51 \pm 0.18$ *and* $\varepsilon_{\mathrm{reach}}/\varepsilon_{\mathrm{crit}} = 8.72 \pm 1.03$*. Empirically* $\varepsilon_{\mathrm{reach}} \approx \varepsilon_{\mathrm{spec}}/a$ *with* a *the ReLU active fraction:* $a \approx 0.6$ *gives* $1/a \approx 1.5$–1.7*, matching the measured* $\varepsilon_{\mathrm{reach}}/\varepsilon_{\mathrm{spec}}$*, and composing with the bracket gives the observed envelope* $\varepsilon_{\mathrm{reach}}/\varepsilon_{\mathrm{crit}} \lesssim C/a$*, spanning roughly* $[2, 10]$*.*

This is a conjecture; two independent proof gaps remain open. *(1)* Masked-operator spectral scaling (unproved): *the relation* $\varepsilon_{\mathrm{reach}} \approx \varepsilon_{\mathrm{spec}}/a$ *assumes the dominant eigenvalue of the masked Jacobian* $D_a(\hat{A}' \otimes W)$ *scales as* $\approx a\,\rho(\hat{A}')\rho(W)$*, but no such lemma holds in general because the active set correlates with the equilibrium and with* u_1*; this is a mean-field heuristic calibrated to two operating points. (2)* Linear-to-nonlinear bifurcation (empirical only): *the step from "a linearized eigenvalue reaches 1" to "the true nonlinear fixed point destabilizes" is supported here only empirically across the 10 seeds (a clean simple-pole divergence, critical exponent* $\gamma \approx 1.02$–1.06*). We present* $\varepsilon_{\mathrm{reach}} \approx \varepsilon_{\mathrm{spec}}/a$ *as an empirical regularity organized by*

Table 7: Reconciliation of the robustness constants. The three "×" ratios that recur in the text are distinct quantities, not estimates of one number: the operating margin is how far the trained model sits inside $\rho = 1$, the empirical conservatism is how far $\varepsilon_{\text{crit}}$ understates the true nonlinear break, and the proven bracket is the worst-case enclosure constant.

Quantity	Meaning	Source
$\kappa = \lVert J_z \rVert_2$	trained contractivity (< 1)	table 1
$\varepsilon_{\text{crit}} = \frac{1-\kappa}{\lVert W \rVert_2}$	norm certificate (safe radius)	Theorem 1
$\varepsilon_{\text{spec}}$	all-active spectral break budget	Theorem 2
2–4×	operating ρ-margin to $\rho = 1$	appendix F.4
2–9×	empirical $\varepsilon_{\text{reach}}/\varepsilon_{\text{crit}}$ (10 seeds)	Remark 2
10–16×	proven bracket constant C/β	Theorem 2
$\gamma \approx 1.02$	resolvent critical exponent	Remark 2
$g_W \in [1.19, 2.47]$	nonnormality $\lVert W \rVert_2 / \rho(W)$	Theorem 2

the same two knobs (g_W and a), not a proven extension. The practical consequence is benign: $\varepsilon_{\text{reach}}$ runs 5–11× the largest realistic budget ($\varepsilon \le 0.2$), so realistic-budget robustness is governed by the first-order $\sigma_1(S_c)$ of fig. 7, not by proximity to criticality.

E The AEGIS-Conformal Coverage Bound

This section derives the certificate behind eq. (8): a worst-case conformity-score-shift bound $L_1^c\varepsilon + C_v\varepsilon^2$ and the robust-coverage guarantee it implies. The argument has three parts, each building on the sensitivity already established. Definition 2 fixes the two conformity scores and the two constants. Lemma 1 bounds the worst-case score shift using the first-order margin sensitivity (the Cauchy–Schwarz step of Proposition 2) and the second-order curvature (the complete remainder of section C.1). Theorem 4 inflates the split-conformal threshold by that shift and reduces to the binary certificate of Zargarbashi, Antonelli, and Bojchevski (2023). We use (A1)–(A3) and operate in the certified regime $\varepsilon < \varepsilon_{\text{crit}}$.

Conformal setup. Fix a target node v with true label $y_v \in [C]$, and let $f = Wz_v^* + b \in \mathbb{R}^C$ be its pre-softmax logits (rows $W_1, \dots, W_C$) with class probabilities $\pi = \mathrm{softmax}(f)$. For a feasible perturbation $\delta\hat{A}$ of size at most ε, write the perturbed logits $f' = W{z^*}'_v + b$ and probabilities $\pi' = \mathrm{softmax}(f')$, and let $S_{c,v}$ be the block rows of S_c at node v. By the first-order sensitivity (14),

$$\Delta z_v^* = S_{c,v}\,\boldsymbol{\delta} + R_v, \qquad \lVert\boldsymbol{\delta}\rVert_2 = \left\lVert \delta\hat{A} \right\rVert_F \le \varepsilon, \tag{33}$$

where $\boldsymbol{\delta}$ are the edge-basis coordinates of $\delta\hat{A}$ and R_v is the remainder bounded in section C.1. The edge basis is orthonormal, so the coordinate map is an isometry and the Frobenius budget passes through unchanged, which is the normalization that makes $\sigma_1(S_c)$ consistent with $\left\lVert \delta\hat{A} \right\rVert_F$.

Definition 2 (Conformity scores and the two AEGIS constants)**.** *For node v and candidate label $r \in [C]$, define the two conformity scores (higher means more conforming; the prediction set is $C(v) = \{r : \mathrm{score}_r(v) \ge \hat{q}\}$):*

$$\textit{TPS:} \quad \mathrm{score}_r^{\mathrm{TPS}}(v) = \pi_r, \tag{34}$$

$$\textit{APS:} \quad \mathrm{score}_r^{\mathrm{APS}}(v) = 1 - \big(\rho_r + u_v\,\pi_r\big), \tag{35}$$

$$\rho_r := \sum_{c:\,\pi_c > \pi_r} \pi_c, \tag{36}$$

with $u_v \sim \mathrm{Unif}[0,1]$ a per-node tie-break drawn once and shared between calibration and test, so it cancels under exchangeability. For each competitor $c \ne r$ let $g_c := f_r - f_c$ be the logit margin. Define the first-order and curvature constants

$$L_1^{(c)} := \lVert (W_r - W_c)\, S_{c,v} \rVert_2\,, \tag{37}$$

$$C_v := \lVert W_r - W_c \rVert_2 \cdot \frac{L_{J,v}}{2(1-\kappa)^2}, \tag{38}$$

where $L_{J,v} \le \lVert W \rVert_2^2 \lVert z^* \rVert$ is the per-node curvature of section C.1 and $\kappa = \lVert J_z \rVert_2 < 1$ (A3). Equation (37) is the first-order margin sensitivity; (38) is the curvature constant $L_J/(2(1-\kappa)^2)$ of Proposition 3 scaled by the readout gap. The aggregate node constant in eq. (8) takes the competitor-worst values, $L_1^c = \max_{c \ne r} L_1^{(c)}$ and C_v with the worst $\lVert W_r - W_c \rVert_2$.

Remark 3 (Why the gradient is the readout gap, not $\nabla_z\mathrm{score}$)**.** *The constant (37) uses the logit-margin rows $W_r - W_c$ rather than a softmax-Jacobian row $\nabla_z\mathrm{score}_r$, for two reasons. First, both scores depend on the logits only through the margins $\{g_c\}_{c \ne r}$ (softmax sees logits only through differences), so controlling every margin controls π_r and ρ_r and hence both scores (Lemma 1). Second, the margin form needs no bound on softmax curvature: it yields a sound, one-sided score drop without the $1/4$ softmax-Lipschitz slack. The $\nabla_z\mathrm{score}$ form is literally correct only for TPS, whereas the margin form covers both.*

Lemma 1 (Worst-case conformity-score shift)**.** *Assume (A1)–(A3), $\varepsilon < \varepsilon_{\text{crit}}$, and a linear head $f = Wz_v^* + b$. Then for every feasible $\delta\hat{A}$ with $\left\lVert \delta\hat{A} \right\rVert_F \le \varepsilon$ and every label r,*

$$\big| g_c(\hat{A} + \delta\hat{A}) - g_c(\hat{A}) \big| \le L_1^{(c)}\varepsilon + C_v\varepsilon^2 \tag{39}$$

for every competitor $c \ne r$, and consequently the worst-case drop of the conformity score of label r obeys

$$\mathrm{score}_r(v;\hat{A}) - \mathrm{score}_r(v;\hat{A} + \delta\hat{A}) \le \Delta_r(\varepsilon),$$
$$\Delta_r(\varepsilon) := \Psi_r\big(\{L_1^{(c)}\varepsilon + C_v\varepsilon^2\}_{c \ne r}\big), \tag{40}$$

where Ψ_r is the monotone map from margin decrements to the induced score decrement. Taking the competitor-worst constants gives the headline $\Delta_r(\varepsilon) \le L_1^c\varepsilon + C_v\varepsilon^2$.

Proof. Step 1: the linear term (Cauchy–Schwarz). Because the head is affine, $g_c = (W_r - W_c)z_v^* + (b_r - b_c)$, so $g'_c - g_c = (W_r - W_c)\Delta z_v^*$. Inserting (33) splits this into a first-order and a remainder part,

$$g'_c - g_c = (W_r - W_c)\, S_{c,v}\,\boldsymbol{\delta} + (W_r - W_c)\, R_v. \tag{41}$$

The first summand is a fixed covector paired with $\boldsymbol{\delta}$, so Cauchy–Schwarz and $\lVert\boldsymbol{\delta}\rVert_2 \le \varepsilon$ bound it by $L_1^{(c)}\varepsilon$ through definition (37). This is the same step that proves the radius r_v, now read with the readout gap $W_r - W_c$ in place of the full S_v image, so the constant is the rate at which the most exposed margin moves per unit perturbation.

Step 2: the quadratic term (curvature). For the remainder, $|(W_r - W_c)R_v| \le \|W_r - W_c\|_2 \|R_v\|_2$. By section C.1 the equilibrium-map Hessian is the complete second derivative (22), bounded in operator norm by $(1-\kappa)^{-2}L_{J,v}$ with $L_{J,v} \le \|W\|_2^2 \|z^*\|$ finite under the contraction (A3). A second-order Taylor expansion along the feasible segment (finitely many ReLU regions, conservative IFT on each) therefore gives $\|R_v\|_2 \le \frac{1}{2}(1-\kappa)^{-2}L_{J,v}\,\varepsilon^2$, hence $|(W_r - W_c)R_v| \le C_v\varepsilon^2$ by (38). Combining the two summands in (41) with the triangle inequality yields (39).

Step 3: from margins to scores (monotone link). Both scores see the logits only through the margins: $\pi_r = (\sum_c e^{-g_c})^{-1}$ with $g_r := 0$, and the rank events $\{\pi_c > \pi_r\} = \{g_c < 0\}$ defining ρ_r depend only on the signs of the margins. The adversary minimizing score_r therefore lowers every margin as far as Step 1–2 allow, at most $L_1^{(c)}\varepsilon + C_v\varepsilon^2$ each. Define Ψ_r as the resulting worst-case score drop. For TPS, π_r is coordinatewise nonincreasing in each margin decrement, so substituting the worst admissible decrement per competitor and recomputing the softmax gives a sound over-estimate of the drop with no softmax-Lipschitz constant needed. For APS, lowering a margin can only raise score_r through the $-u_v\pi_r$ term (favorable) or move a competitor above r in rank, adding π_c to ρ_r (adverse); the worst case uses the same lowered-margin softmax, and since $\rho_r, \pi_r \in [0,1]$ and $u_v \le 1$ the resulting drop is exact rather than a Lipschitz bound. Either way the per-competitor decrement caps the per-label score drop, giving (40); the competitor-worst constants give the headline form. □

The shift bound is a margin budget read in two pieces. The linear constant L_1^c is the rate at which the most exposed decision margin moves, factored as the readout gap $\|W_{y_v} - W_c\|_2$ times the equilibrium sensitivity $S_{c,v}$, the same two-amplifier product as the radius r_v but expressed in score units. The quadratic constant C_v is the price of leaving the linear regime, the curvature $L_J/(2(1-\kappa)^2)$ with its two resolvents, scaled by the readout gap. Carrying the quadratic term is what makes the certificate sound rather than merely first-order: the bare $\sigma_1(S_c)\varepsilon$ screen can be breached just past r_v, whereas $L_1^c\varepsilon + C_v\varepsilon^2$ over-states the true drop for every $\varepsilon < \varepsilon_{\mathrm{crit}}$, which is why the empirical gate is conservative (0.92–0.98 at $\varepsilon = 0.05$) rather than tight.

Remark 4 (Where rigor stops, honestly)**.** *Two load-bearing conditions remain. (1)* ***Affine head.*** *Step 1 needs f affine in z_v^*; a nonlinear readout would replace $W_r - W_c$ by the head Jacobian and add a Lipschitz factor. Our models use a linear head, so this holds with equality. (2)* ***Contractive regime.*** *The curvature bound and the finiteness of $\|z^*\|$ both need $\kappa < 1$ and $\varepsilon < \varepsilon_{\mathrm{crit}}$ (A3); past $\varepsilon_{\mathrm{crit}}$ the resolvent $1/(1-\kappa)$ is meaningless and the bound is void, the same scope as Theorem 1. On the conformal subgraph $\kappa \approx 0.68$, so $(1-\kappa)^{-2} \approx 9.8$ inflates C_v, yet the gate confirms the bound is not breached. The earlier worry about "omitted higher-order terms" does not arise: section C.1 shows the operator is bilinear on each linear region, so* (22) *is the* complete *equilibrium Hessian and C_v rests on the full second derivative, not a dominant-term reduction.*

Theorem 4 (Robust coverage of AEGIS-Conformal)**.** *Let $\{(v_i, y_{v_i})\}_{i\in\mathrm{cal}}$ be a split-conformal calibration set and v a test node. Assume:*

- *(C1)* Exchangeability. *The clean true-label scores $\{\mathrm{score}_{y_{v_i}}(v_i)\}_{i\in\mathrm{cal}} \cup \{\mathrm{score}_{y_v}(v)\}$ are exchangeable (an inductive split, or a transductive split with a permutation-invariant predictor; Zargarbashi, Antonelli, and Bojchevski 2023).*
- *(C2)* Score-shift bound. *For every calibration node w and the test node, simultaneously over all feasible $\left\|\delta\hat{A}\right\|_F \le \varepsilon < \varepsilon_{\mathrm{crit}}$, $\mathrm{score}_{y_w}(w; \hat{A} + \delta\hat{A}) \ge \mathrm{score}_{y_w}(w; \hat{A}) - \Delta_{y_w}(\varepsilon)$ with Δ_{y_w} the bound of Lemma 1.*

Form the calibration-robust threshold

$$\hat{q}_{\mathrm{rob}} = \mathrm{Quantile}_{\lceil (n_{\mathrm{cal}}+1)(1-\alpha)\rceil / n_{\mathrm{cal}}} \Big(\big\{\mathrm{score}_{y_{v_i}}(v_i; \hat{A}) - \Delta_{y_{v_i}}(\varepsilon)\big\}_{i\in\mathrm{cal}}\Big), \quad (42)$$

and the robust set $C_\varepsilon(v; \hat{A}') = \{r : \mathrm{score}_r(v; \hat{A}') \ge \hat{q}_{\mathrm{rob}}\}$. Then

$$\Pr\big[y_v \in C_\varepsilon(v; \hat{A} + \delta\hat{A})\big] \ge 1 - \alpha \quad \textit{for all feasible } \left\|\delta\hat{A}\right\|_F \le \varepsilon. \quad (43)$$

Proof. Step 1: reduce to lowered scores. Define the lowered scores $\tilde{\mathrm{score}}_{y_w}(w) := \mathrm{score}_{y_w}(w; \hat{A}) - \Delta_{y_w}(\varepsilon)$ on calibration and test. By (C2), the realized perturbed true-label score dominates its lowered clean value, $\mathrm{score}_{y_w}(w; \hat{A} + \delta\hat{A}) \ge \tilde{\mathrm{score}}_{y_w}(w)$, for every feasible $\delta\hat{A}$ at once. This is exactly the guaranteed lower envelope the binary certificate of Zargarbashi, Antonelli, and Bojchevski (2023) requires; Lemma 1 supplies it analytically for the continuous ε-ball, replacing their combinatorial enumeration of a discrete neighbourhood.

Step 2: the lowered scores stay exchangeable. Each node's lowering $\Delta_{y_w}(\varepsilon)$ is a deterministic function of that node's own $(W_{y_w} - W_c,\ S_{c,w},\ L_{J,w})$, a per-point measurable transform of its clean score context, and the tie-break $u_{(\cdot)}$ is shared. Under (C1) the clean scores are exchangeable, and a common per-point transform preserves exchangeability, so the lowered scores are exchangeable too.

Step 3: split-conformal validity on the lowered scores. By the standard split-conformal guarantee (Vovk, Gammerman, and Shafer 2005) applied to the exchangeable lowered scores, with $\hat{q}_{\mathrm{rob}}$ the $\lceil (n_{\mathrm{cal}}+1)(1-\alpha)\rceil$-th smallest calibration value (42), $\Pr[\tilde{\mathrm{score}}_{y_v}(v) \ge \hat{q}_{\mathrm{rob}}] \ge 1 - \alpha$.

Step 4: transfer to the whole ball. On the event $\{\tilde{\mathrm{score}}_{y_v}(v) \ge \hat{q}_{\mathrm{rob}}\}$, Step 1 gives $\mathrm{score}_{y_v}(v; \hat{A} + \delta\hat{A}) \ge \tilde{\mathrm{score}}_{y_v}(v) \ge \hat{q}_{\mathrm{rob}}$ for every feasible $\delta\hat{A}$ simultaneously, because the envelope is uniform over the ball rather than per-δ. Hence $y_v \in C_\varepsilon(v; \hat{A} + \delta\hat{A})$ on that event, and (43) follows from the bound of Step 3. At $\varepsilon = 0$, $\Delta \equiv 0$ and the statement reduces to the ordinary split-conformal $1 - \alpha$ guarantee, as it must. $\square$

The certificate buys robustness by paying the worst-case score drop up front, on the calibration set. Lowering every calibration true-label score by its own analytic $\Delta(\varepsilon)$ before taking the quantile raises the threshold just enough that any in-ball perturbation of the test node, which can lower its true-label score by at most its own $\Delta(\varepsilon)$, still clears the bar. Because the lowering is per-node and deterministic it commutes with exchangeability, so the only probabilistic content is the ordinary split-conformal quantile event; the uniformity in Step 4 is what upgrades "robust to a fixed attack" into "robust over the whole ball."

Remark 5 (Honest status of the coverage guarantee)**.** *(C1) is the load-bearing hypothesis, and it does not hold for free on a single fixed transductive graph: calibration and test nodes share one realized adjacency, and an IGNN's output at v depends on the whole graph, so the scores are not i.i.d. and exchangeability must be imposed by design (inductive sampling, or a transductive-exchangeability argument as in Zargarbashi, Antonelli, and Bojchevski 2023). We therefore state* (43) *as sound under (C1), with the empirical coverage (clean coverage* ≈ 0.90*) and the worst-case gate (*0.90 *at* $\varepsilon = 0.01$*,* 0.92–0.98 *at* $\varepsilon = 0.05$*, zero equilibrium divergence across all* 4138 *gate nodes) as evidence that (C1) holds well enough on these graphs. The reduction to Zargarbashi, Antonelli, and Bojchevski (2023) is faithful: their construction yields robust coverage from any uniform worst-case score envelope, and Lemma 1 is exactly such an envelope, computed analytically with zero Monte-Carlo smoothing.*

F Detailed Experimental Results

This section gives the detailed tables and per-dataset breakdowns behind the headline numbers of section 5, followed by the fraud-detector walk-through and the reproducibility note. All results use the 10 fixed seeds of appendix H.

F.1 Four-Quadrant Attack Comparison

Table 2 evaluates AEGIS across the four quadrants of (gradient-based, gradient-free) $\times$ (equilibrium-shift, classification-loss). The SVD direction maximizes first-order equilibrium shift by construction (Proposition 1); what matters is non-circularity and cost. A transfer direction from a separately trained surrogate (zero shared gradients) recovers 99% of the one-query damage ($\cos = 0.99$), versus $44 \pm 4\%$ for a 512-query black-box search, so the fault line is model-intrinsic rather than a gradient artifact. Per query it delivers 74–$156\times$ the damage of a 50-step PGD (Madry et al. 2018); classification-loss PGD inflicts 15–70% less at $50\times$ the cost, IFT-gradient PGD (solver validation, not an independent baseline) 72–92%, and prediction flips stay 0–1.8% for all methods. Figure 8 visualizes the equilibrium-damage column.

Radius vs. smoothing. Randomized smoothing (Bojchevski, Klicpera, and Günnemann 2020) gives constant

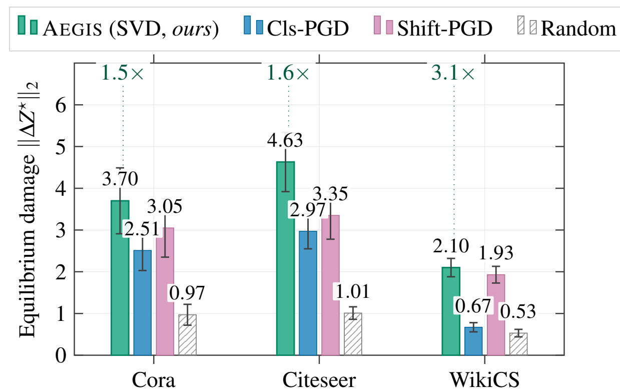

Figure 8: Equilibrium damage $\|\Delta Z^{\star}\|_2$ at ε=0.10 across 10 seeds, for the one-query AEGIS (SVD) attack against classification-loss PGD, IFT-gradient (shift) PGD, and a random baseline, on Cora, Citeseer, and WikiCS. Visual companion to table 2: one analytic AEGIS query inflicts 1.5–$3.1\times$ the damage of a 50-step Cls-PGD run per dataset, at a fraction of its cost.

per-node base radii (0.123 vs. AEGIS's 0.078 at σ=0.05), and localized smoothing (Schuchardt et al. 2023) needs 10^3–10^4 MC samples per node; both lack edge structure. AEGIS radii are instead structurally informative ($r_v \approx 0.10$ in dense regions, 0.01 at boundaries), a complementary point on the certificate-versus-diagnostic frontier.

Classification-gradient edges differ. Per-edge classification gradients anti-correlate with discrete ground truth on Cora ($\tau = -0.20_{\pm 0.05}$ vs. AEGIS's $+0.32_{\pm 0.04}$) and stay near zero elsewhere; AEGIS achieves 2–$6\times$ higher P@10, confirming that equilibrium and classification sensitivities are distinct vulnerability surfaces. Per-edge finite differences reproduce the S_c column-norm ranking (τ=0.999), and iterative S_c recomputation closes ≤ 3 pp of the static-versus-greedy gap, so the static ranking is the headline. The four-quadrant equilibrium-damage table itself is promoted to the main text (table 2, section 5.1); the per-edge analyses here are the supporting detail.

F.2 Comparison with Prior Structural Baselines

Table 8 reports head-to-head numbers against the GR-BCD and PR-BCD structural attackers (Geisler et al. 2021). AEGIS is a label-free one-query proxy; the question is how much of each gold-standard signal it retains. GR-BCD aligns with S_c on Pubmed ($\tau = +0.69$) but decorrelates on Cora ($\tau = +0.16$); the budgets shown ($k \leq 10$) are AEGIS's early-warning regime, where its one-pass diagnostic is competitive, while GR-BCD/PR-BCD dominate raw damage at the high budgets they target. AGNNCert (Li and Wang 2025) is a sound certificate against a bounded *number* of discrete edge edits (a majority vote over hash-partitioned sub-graphs), whereas AEGIS's r_v bounds a *continuous* edge-weight budget $\left\|\delta \hat{A}\right\|_F$; the two target different threat models and are complementary rather than comparable on one radius axis. AEGIS also beats degree-proportional, edge-betweenness, and spectral rankings at ε=0.01 (+6–148% AtkAdv, Wilcoxon $p<0.001$) and inflicts 3–$10\times$ more equilibrium damage than Met-

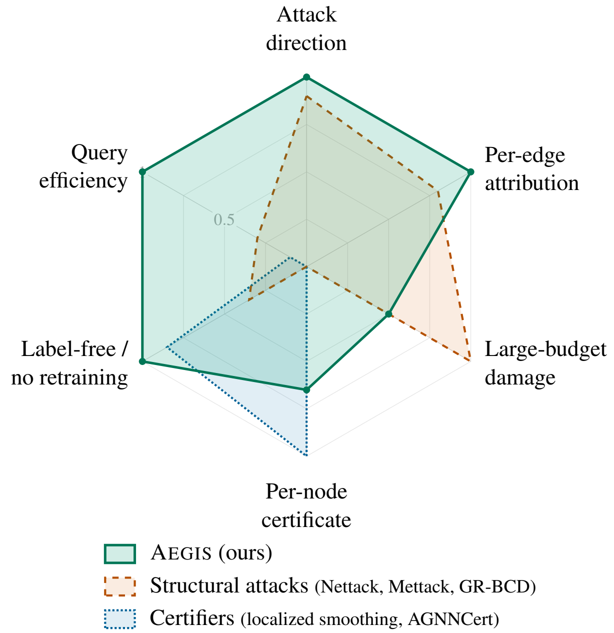

Figure 9: Method-positioning radar (axis levels qualitative; 0 at the centre, 1 at the outer ring). Structural attacks (Zügner, Akbarnejad, and Günnemann 2018; Zügner and Günnemann 2019a; Geisler et al. 2021) surface an attack direction and per-edge ranks and dominate raw damage at large budgets, but give no per-node certificate; certifiers (Schuchardt et al. 2023; Li and Wang 2025) give sound per-node radii but name no vulnerability-driving edges; AEGIS reads attack direction, per-edge attribution, and a first-order per-node certificate off a single closed-form S_c query, label-free and at far lower cost. The specialists peak on their own axis; AEGIS is the unified option covering all six.

Table 8: AEGIS vs. external baselines (10 seeds). GR-BCD (Geisler et al. 2021) is the BCD-family scalable structural attacker (PR-BCD is the same paper's larger-budget variant, see text).

Baseline	Setting	AEGIS	Baseline	τ
GR-BCD	Pubmed k=10	0.356	0.350	**+0.69**
GR-BCD	Cora k=5	0.643	1.207	+0.16

tack (Zügner and Günnemann 2019a) in the early-warning regime $k \in \{1, \ldots, 5\}$ ($\mathbf{149/150}$ paired wins, $p{<}10^{-43}$).

Figure 9 positions these methods on a capability radar. No single method covers every axis: structural attacks win raw damage at large budgets, certifiers win the sound per-node certificate, and AEGIS is the only one that reads an attack direction, per-edge attribution, and a first-order per-node certificate off a single closed-form, label-free S_c query.

F.3 Certification vs Randomized Smoothing

The matched smoothing $\sigma{=}\varepsilon/\sqrt{2|E|}$ (Frobenius ball) is so small that the smoothed classifier is unchanged and abstains; only on the strictly larger per-coordinate ball ($\sigma{=}\varepsilon$) does it certify (0.77–0.96 of nodes), and then at $\mathbf{23{,}000\text{–}57{,}000}\times$ the wall-clock of the zero-sample S_c bound. Even on the matched ball the analytic certificate is non-vacuous and $\mathbf{11{,}700\text{–}16{,}700}\times$ faster: the comparison favors smoothing twice (larger ball, best matching), yet the certificate of table 4 dominates by four orders of magnitude.

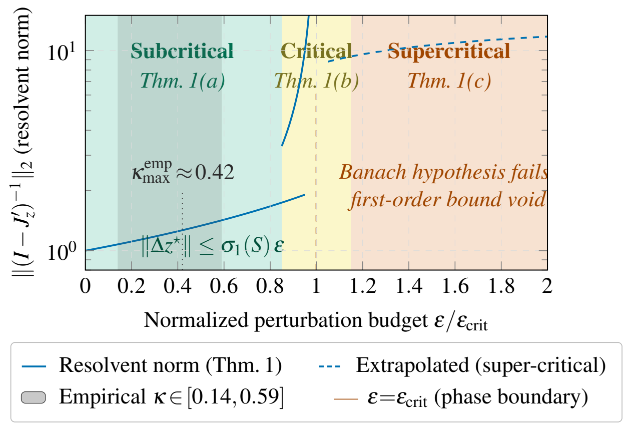

Figure 10: Three-regime characterization of Theorem 1. The empirical κ band (grey) saturates well below the spectral-normalization ceiling, so the empirical resolvent growth is far milder than the worst-case $1/(1-\kappa)$ divergence at $\varepsilon \to \varepsilon_{\text{crit}}$.

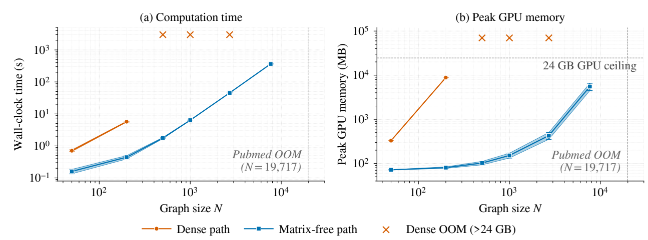

Figure 11: Dense vs. matrix-free AEGIS pipeline (RTX 4090). The dense path scales as $O((Nd)^3)$, exceeding 24 GB beyond N=200; the matrix-free path is roughly $O(N \log N)$ in time and sub-linear in memory, reaching Amazon Photo (N=7,650) at 365 s and 5.5 GB.

F.4 Phase Transition and Scalability

Phase transition. Figure 10 sweeps $\kappa_{\max}{\in}[0.30, 0.99]$ on Cora (10 seeds). Even when the spectral-normalization cap is pushed to 0.99, driving the theoretical $\varepsilon_{\text{crit}}$ down 230×, the trained ReLU pattern keeps $\rho(J_z) \le 0.42$, so the resolvent grows only $\mathbf{1.17{\to}1.80}$. This is a **2–4**× spectral-radius margin to criticality (ρ=1), the operative quantity behind $\varepsilon_{\text{crit}}$, stable across the sweep. Training converges throughout, and the matrix-free pipeline reaches its 50-step fixed-point budget at $\kappa_{\max}{\ge}0.85$.

Scalability and matrix-free self-consistency. Figure 11 compares the dense and matrix-free pipelines. The matrix-free $S_c v$ implements the dense S_c up to Neumann truncation; σ_1 agrees within 0.03% at N=200 (per-edge τ=0.999), and the analytic truncation residual $\kappa^{200} \in [10^{-105}, 10^{-48}]$ across the suite (eq. (13)). For larger N the rSVD error (Halko, Martinsson, and Tropp 2011) is bounded by the spectral gap; Pubmed (N=19,717) exceeds 24 GB on the dense path.

F.5 Ablations and Defense

Subgraph size. Varying $N{\in}\{30, 50, 100, 200\}$ on Cora (10 seeds), tightness is stable at 1.01–1.02 for $N{\le}100$ and degrades to 1.031 at N=200; we use N=50 by default

Table 9: S_c framework on 7 GNN architectures (Cora, 10 seeds). Tight.: actual/predicted shift ratio. AtkAdv: AEGIS/random damage. τ: Kendall of the *unweighted* v_{ij} ranking vs. single-edge removal; the edge-weighted $A_{ij}v_{ij}$ ranking of Proposition 3 (fig. 5) is the higher headline score. Acc: full-graph test accuracy.

Model	Tight.	AtkAdv	τ	Acc
IGNN (eq.)	$1.01_{\pm.00}$	$7.6_{\pm0.5}\times$	$+.32_{\pm.04}$	$77.5_{\pm1.7}$
GCN-2	$1.00_{\pm.00}$	$3.6_{\pm0.6}\times$	$-.04_{\pm.03}$	$78.9_{\pm0.9}$
GCN-4	$1.02_{\pm.00}$	$3.4_{\pm0.4}\times$	$+.49_{\pm.02}$	$78.3_{\pm1.8}$
GIN-2	$1.01_{\pm.00}$	$2.6_{\pm0.1}\times$	$+.33_{\pm.03}$	$76.6_{\pm1.9}$
GAT†	$1.01_{\pm.01}$	$2.1_{\pm0.1}\times$	$+\mathbf{.54}_{\pm.06}$	$77.8_{\pm1.2}$
SAGE-2	$1.00_{\pm.00}$	$1.9_{\pm0.2}\times$	$+.22_{\pm.10}$	$77.5_{\pm1.5}$
APPNP	$1.02_{\pm.00}$	$3.4_{\pm0.3}\times$	$+.35_{\pm.01}$	$\mathbf{82.2}_{\pm0.6}$

(66× faster). **Full-graph scale.** A 50-node BFS covers only $\sim$1.8% of a citation graph's edges (Cora τ=0.16), so at this scale we run AEGIS on the full graph, where its edge advantage over degree-proportional ranking *amplifies* to $\mathbf{9.82}\times$ (Citeseer) and $\mathbf{3.25}\times$ (Cora) at k=10, against $\approx$1.1× on subgraphs. **Hyperparameters.** Tightness is stable across $d\in\{16, 32, 64, 128\}$; the spectral-norm ceiling $c\in\{0.5, 0.9\}$ traces the accuracy–robustness frontier (r_v=0.147 at 72.1% vs. 0.089 at 80.6%).

Edge protection and the delocalization of vulnerability. On the full graph (Cora, N=2,708, 10 seeds), masking the top-k edges by v_{ij} cuts $\sigma_1(S_c)$ damage by 2.4±1.8% at k=5 and 4.6±2.9% at k=10, a **12–46**× gain over random masking but far below the 42–61% a 50-node subgraph reports (where k=5 is already 10% of its edges). The reason is structural: the leading attack mode is *delocalized* (participation ratio 41–89 edges), so removing a handful excises a negligible slice. Reaching subgraph-scale reduction needs 2–8% of edges masked, and adaptive recomputation erodes the gain a further 1–2 pp. AEGIS therefore *localizes* risk for attribution and triage rather than yielding a few-edge defense, the failure mode that sober-look critiques (Mujkanovic et al. 2022) flag.

F.6 Per-Architecture Characterization (Explicit GNNs)

For K-layer explicit GNNs (GCN, SAGE, GIN, APPNP, GAT† (Kipf and Welling 2017; Hamilton, Ying, and Leskovec 2017; Xu et al. 2019b; Klicpera, Bojchevski, and Günnemann 2019; Veličković et al. 2018)) we form S_K=$\partial\text{vec}(Z_K)/\partial\text{vec}(A)$ by finite differences and build S_c from S_K identically (Proposition 4). Table 9 reports the per-architecture characterization on Cora: IGNN carries Theorem 1, while the explicit models receive the computational tool without the regime characterization.

GAT† scope. Standard GAT uses A as a binary mask, so $\partial Z/\partial A_{ij}$=0 on existing edges and S_c is undefined; our GAT† modulates attention by the edge weight ($h'_i = \sum_j \hat{A}_{ij}\alpha_{ij}Wh_j$), restoring differentiability and yielding tightness 0.99, AtkAdv 4.4×, τ=+0.36. Binary-mask architectures (hard attention, max/min aggregation, GATv2 (Brody, Alon, and Yahav 2022)-style dynamic-attention masking) fall outside the framework. **Robust backbones.** Under the $\sigma_1(W)\leq0.9$ cap, RobustGCN-lite (Zhu et al. 2019) and GNNGuard-lite (Zhang and Zitnik 2020) match or exceed IGNN; the cap is a precondition (without it κ>1 voids Theorem 1). **Cross-dataset transfer detail.** The edge-weighted $A_{ij}v_{ij}$ ranking beats a pure edge-weight baseline by $\Delta\tau\approx$+0.25, rising to +0.65 on the fraud graph where uniform weights are uninformative, so v_{ij} adds rank information beyond the weight. GAT-2† is the exception: attention reweights edges, so the unweighted v_{ij} transfers better there (+0.47 vs. +0.35). Proposition 3's sufficient pairwise condition holds for **47–62**% of edge pairs, so the global τ is empirical, not implied; where unweighted v_{ij} is uninformative (GCN-2's near-uniform 2-hop shifts) the weighted ranking recovers the order via the edge weight (−0.28→+0.99, Citeseer).

G Fraud-Detector Audit: Walk-Through

This section expands the audit-in-action summarized in section 6. GNN fraud detectors are a prime adversarial target: a flagged account can try to evade detection by perturbing a few of its behavioural-similarity edges (Zügner, Akbarnejad, and Günnemann 2018; Dou et al. 2020). AEGIS audits such a detector directly, with no labels and no retraining, reading from one S_c query which edges most threaten its predictions.

The fault-line diagram is fig. 6 in section 6: the trained IGNN flags the central account, and the edge-weighted ranking $A_{ij}v_{ij}$ pinpoints the fault lines, reproducing the brute-force single-edge-removal damage ranking from a single S_c query. This auditing claim is what the paper validates at scale, and non-circularly. The ranking transfers across the full fraud graph (fig. 5), precisely where uniform edge weights are uninformative so v_{ij} adds the most rank information ($\Delta\tau$ up to $+\mathbf{0.65}$ over the weight baseline). The SVD direction transfers from a separately trained surrogate (zero shared gradients, cos =0.99; section 5.1), so v_{ij} reflects model-intrinsic vulnerability rather than a gradient artifact; AEGIS out-damages discrete attackers (Mettack, GR-BCD) on their own metric in the early-warning regime (section 5.1 and appendix F.2); and the per-edge v_{ij} scores prioritize which connections to monitor, though adversarial vulnerability itself proves delocalized at scale (appendix F.5). Beyond a smoothing certifier, S_c also names *which* edges and *in which direction* (section 4.1), turning a one-query audit into an actionable defense.

H Reproducibility

All results use 10 fixed random seeds $\{42, 137, 271, 314, 1729, 2718, 3141, 5772, 6561, 9999\}$; reported ± are standard deviations over these seeds. Models are trained in PyTorch (Paszke et al. 2019); IGNN (Gu et al. 2020) uses $F(Z)$=$\text{ReLU}(\hat{A}ZW^\top$+$U(X))$ with $W \in \mathbb{R}^{64\times64}$ spectral-normalized (Miyato et al. 2018) and Adam (learning rate 0.01). Datasets are the public splits of Cora, Citeseer, Pubmed (Sen et al. 2008), Amazon Photo (Shchur et al. 2018), WikiCS (Mernyei and Caron 2020), and Amazon Fraud (Dou et al. 2020). The randomized SVD (Halko, Martinsson, and Tropp 2011) uses k=p=10, n_{iter}=2; the Neumann series early-stops at $\left\|J_z^k b\right\| < 10^{-6}\,\|b\|$. Timings are on a single RTX

4090. Code release follows the disclosure protocol of section 8: the diagnostic-only path (r_v and v_{ij} scores) is released unconditionally, while the attack-direction reconstruction (algorithm 1, steps 3–4) is gated behind institutional-affiliation review.